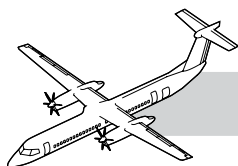


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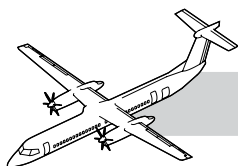
OXYGEN

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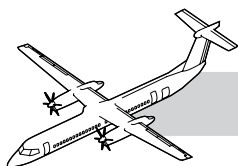
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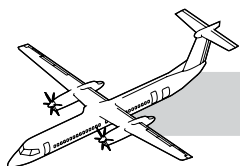
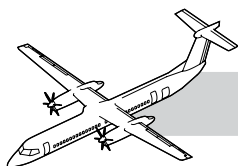


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CHAPTER 35

OXYGEN



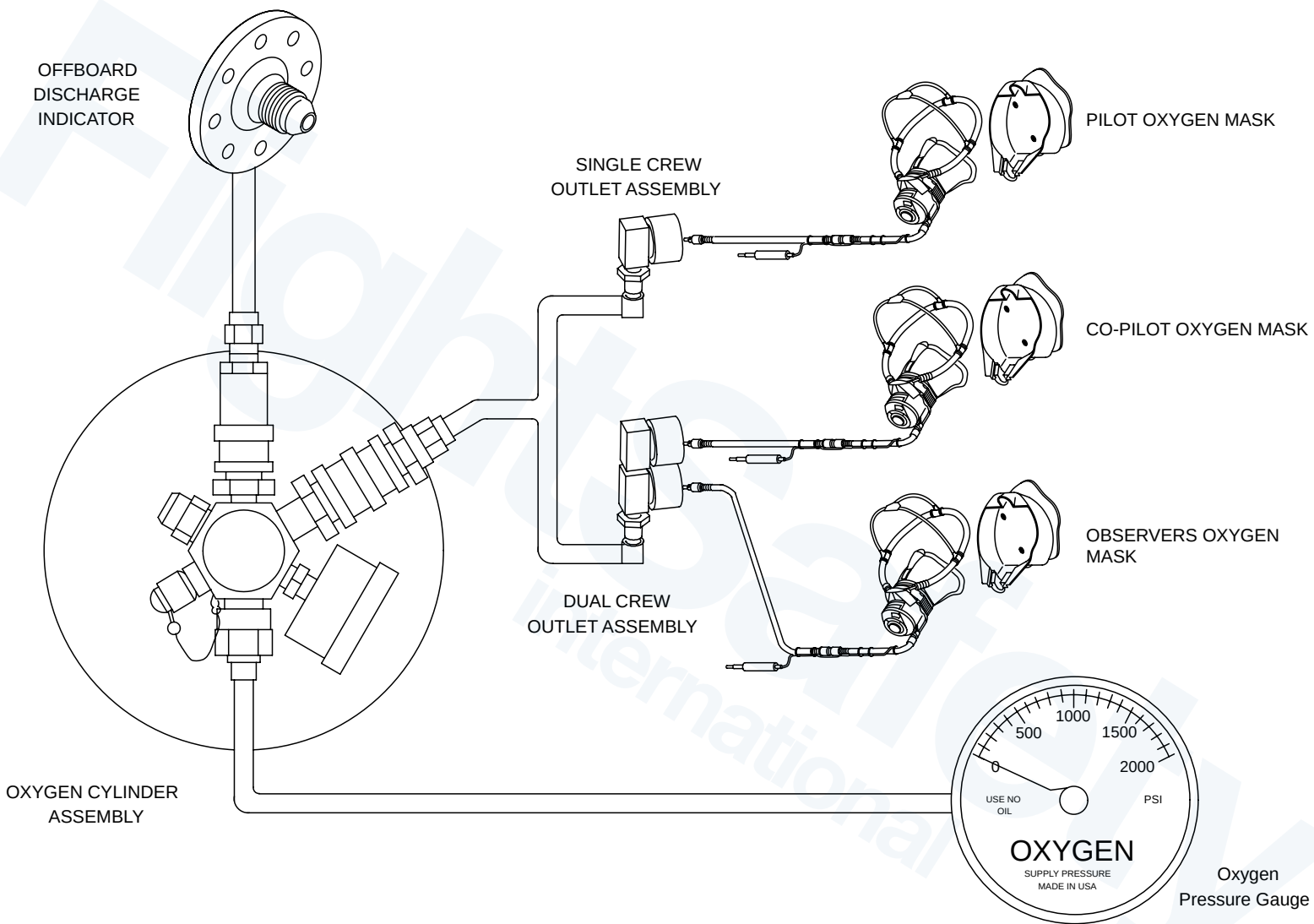
35-00-00 INTRODUCTION

The oxygen system supplies oxygen to the crew and passengers in abnormal or emergency situations.

GENERAL

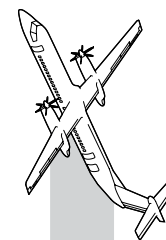
The oxygen system includes the following subsystems:

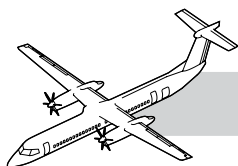
- Crew Oxygen System
- Crew Oxygen Delivery
- Crew Oxygen Indication and Servicing
- Crew Portable Oxygen
- Passenger Oxygen System
- Portable Oxygen System.



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Figure 35-1. Crew Oxygen System





35-10-00 CREW OXYGEN SYSTEM

NOTES

INTRODUCTION

The crew oxygen system supplies oxygen to the crew if there is a loss of pressurization, or there is smoke or fumes in the flight compartment.

GENERAL

Refer to Figure 35-1. Crew Oxygen System.

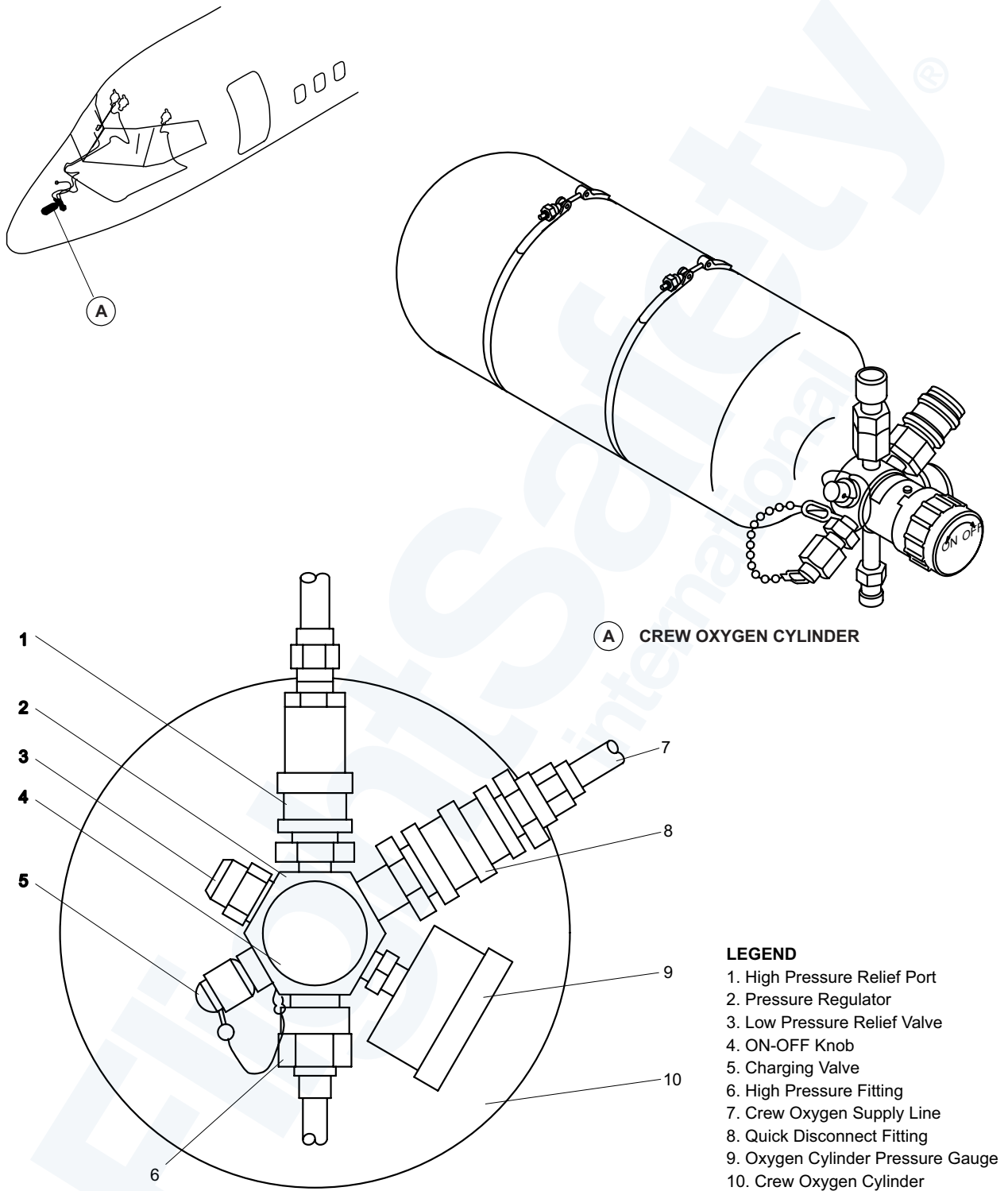
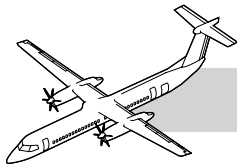
The crew oxygen system is equipped with three oxygen masks: one for the pilot, co-pilot and observer. There is an oxygen cylinder assembly and distribution tubing, along with approved PBE for fire fighting purposes. Smoke goggles are supplied for crew protection in the flight compartment.

The system has a gauge on the co-pilot side console and a second gauge on the oxygen cylinder.

The system includes the following components:

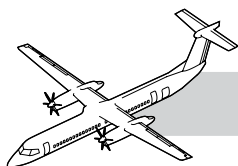
- Oxygen cylinder assembly
- Crew mask and regulator Assemblies
- Overboard discharge indicator
- Outlet assemblies
- Panel mounted pressure gauge
- Low pressure hose assembly
- High pressure hose assembly and
- Capillary line.

The system is capable of supplying enough oxygen for a descent from 25,000 feet to 14,000 feet (7620m to 4267m) in 4 minutes. It is then able to supply oxygen for an additional 116 minutes at 14,000 feet (4267m).



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Figure 35-2. Crew Oxygen Cylinder



COMPONENT DESCRIPTION

Oxygen Cylinder Assembly

Refer to Figure 35-2. Crew Oxygen Cylinder.

The oxygen cylinder, located in the unpressurized lower right nose compartment, is rigidly attached by steel clamps to the outside of the nose landing gear bay side wall.

Fabric curtains isolate the oxygen cylinder assembly from the hydraulic components in the bay immediately aft, and from the hydraulic tubes below.

The capacity of the oxygen cylinder is 39.8 ft³ or 50 ft³ if the optional oxygen cylinder is installed. The rated pressure for both is 1850 psig.

NOTE

The smaller size oxygen cylinder must be removed from the aircraft for servicing, however the larger 50 cu. ft. bottle has provisions for on board servicing.

NOTE

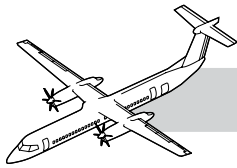
The minimum dispatch pressures for each crew configuration using the 39.8 cu. ft. cylinder is:

- 1383 psi (9536 kPa) for 2 crew
- 1786 psi (12314 kPa) for 3 crew.

The minimum dispatch pressures for each crew configuration using the 50 cu. ft. cylinder is:

- 1050 psi (7239.50 kPa) for 2 crew
- 1450 psi (9997.40 kPa) for 3 crew.

A low pressure regulator supplies a maximum of 300 L/min flow and provides overpressure relief to protect the hoses and masks. The cylinder has a pressure gauge, a quick disconnect fitting, and a high pressure charging valve.



Crew Oxygen Masks

Refer to Figure 35-3. Crew Oxygen Mask.

The quick donning oxygen masks are kept in mask stowage cups attached to the bulkhead above and behind the pilot and co-pilot seats. The observer mask is located on the right hand bulkhead. All three masks are connected to the low pressure distribution lines by individual quick disconnects.

Each mask is equipped with a microphone and an automatic diluter demand regulator. They are designed so they can be worn with the smoke goggles.

Each mask has an approved regulator with the following modes:

- NORM (normal oxygen diluter on-demand)
- 100% (100% oxygen on-demand)
- EMER (emergency pressure oxygen, constant flow).

Mask Stowage Cup

The stowage cups are molded polycarbonate assemblies which help the quick donning requirements.

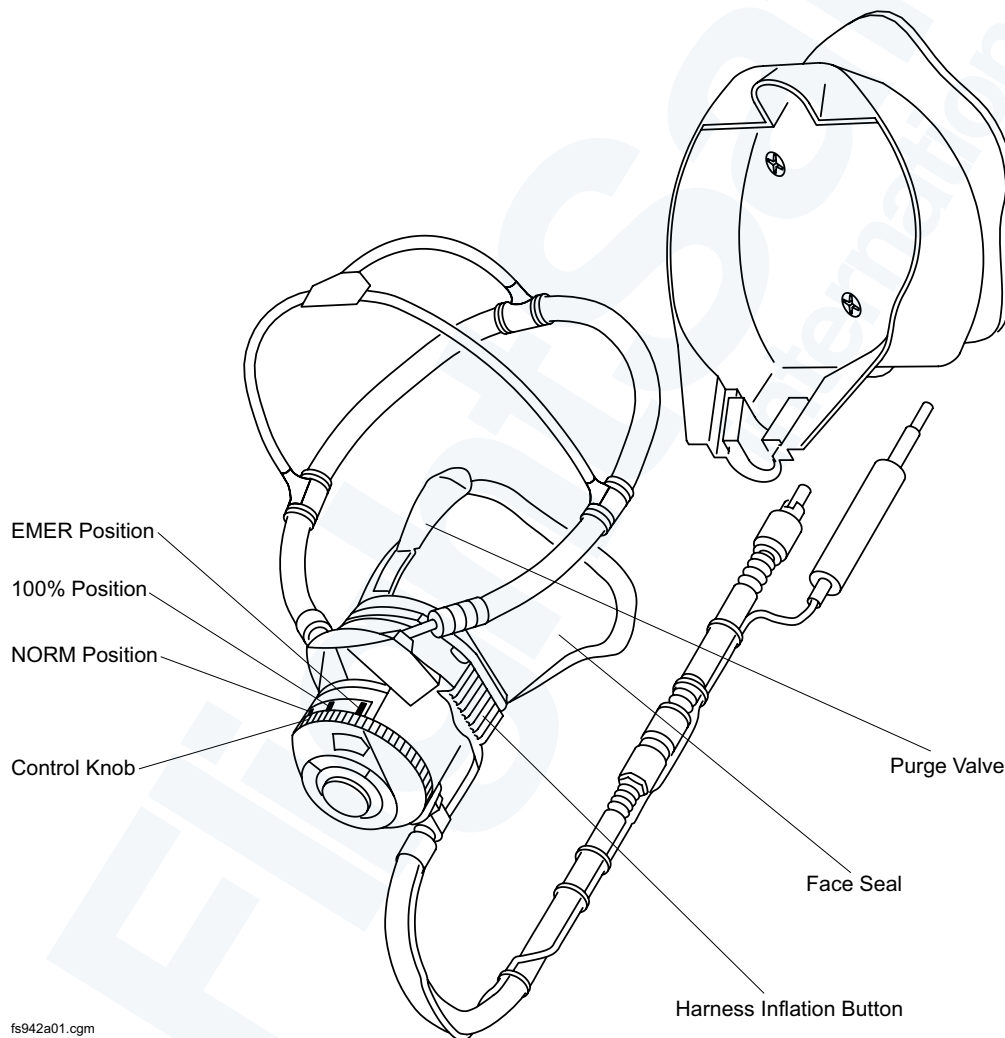
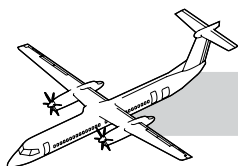


Figure 35-3. Crew Oxygen Mask

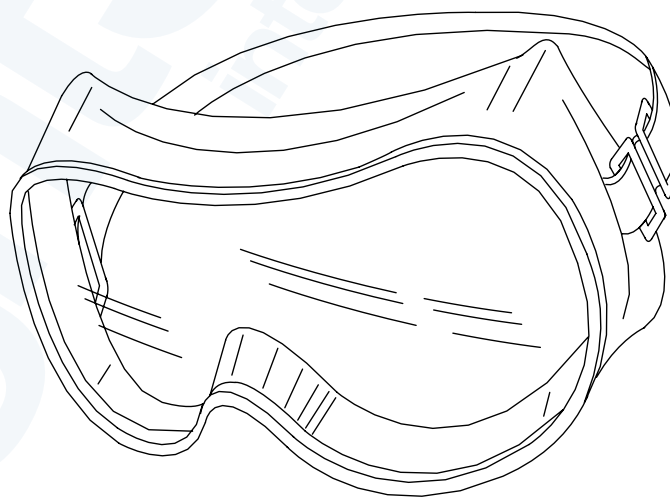
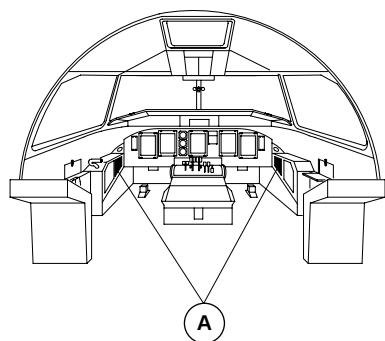


Smoke Goggles

Refer to Figure 35-4. Smoke Goggles.

The smoke goggles are kept in pockets in each side console and at the observer's station.

The smoke goggles have a lexan lens and a silicone face seal and strap.

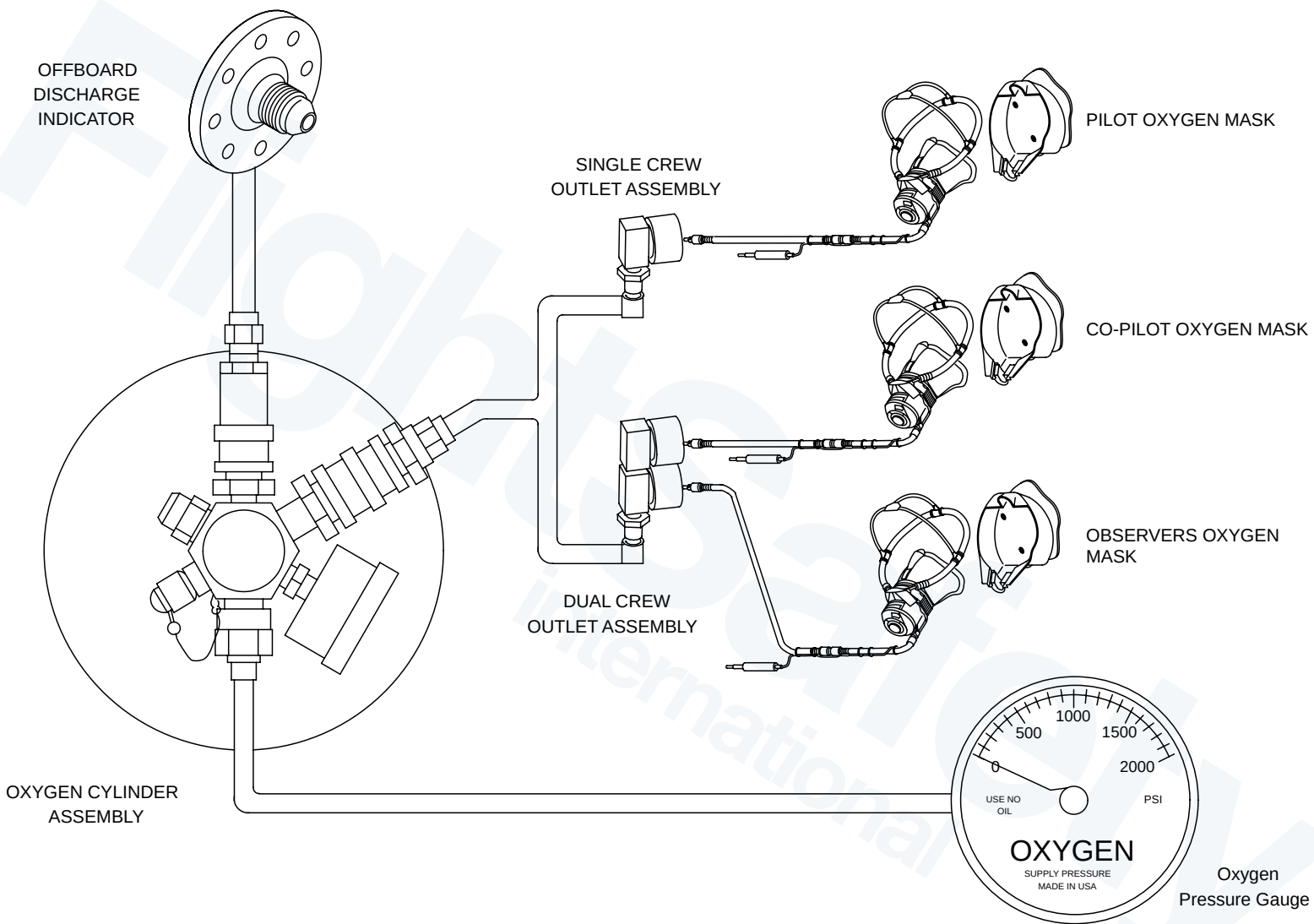


NOTE
One pair shown.
Other pair similar.

A SMOKE GOGGLES

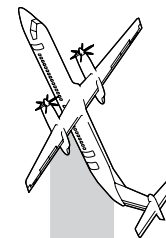
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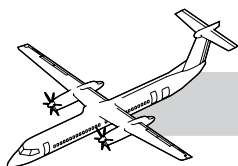
Figure 35-4. Smoke Goggles



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Figure 35-5. Crew Oxygen System





High Pressure Hose Assembly

NOTES

Refer to Figure 35-5. Crew Oxygen System.

The high pressure hose assembly connects the overpressure safety port of the cylinder assembly to the overboard discharge indicator. The hose has an outer braid of stainless steel and is rated to 12,000 psig (82,737 kPag).

Capillary Line

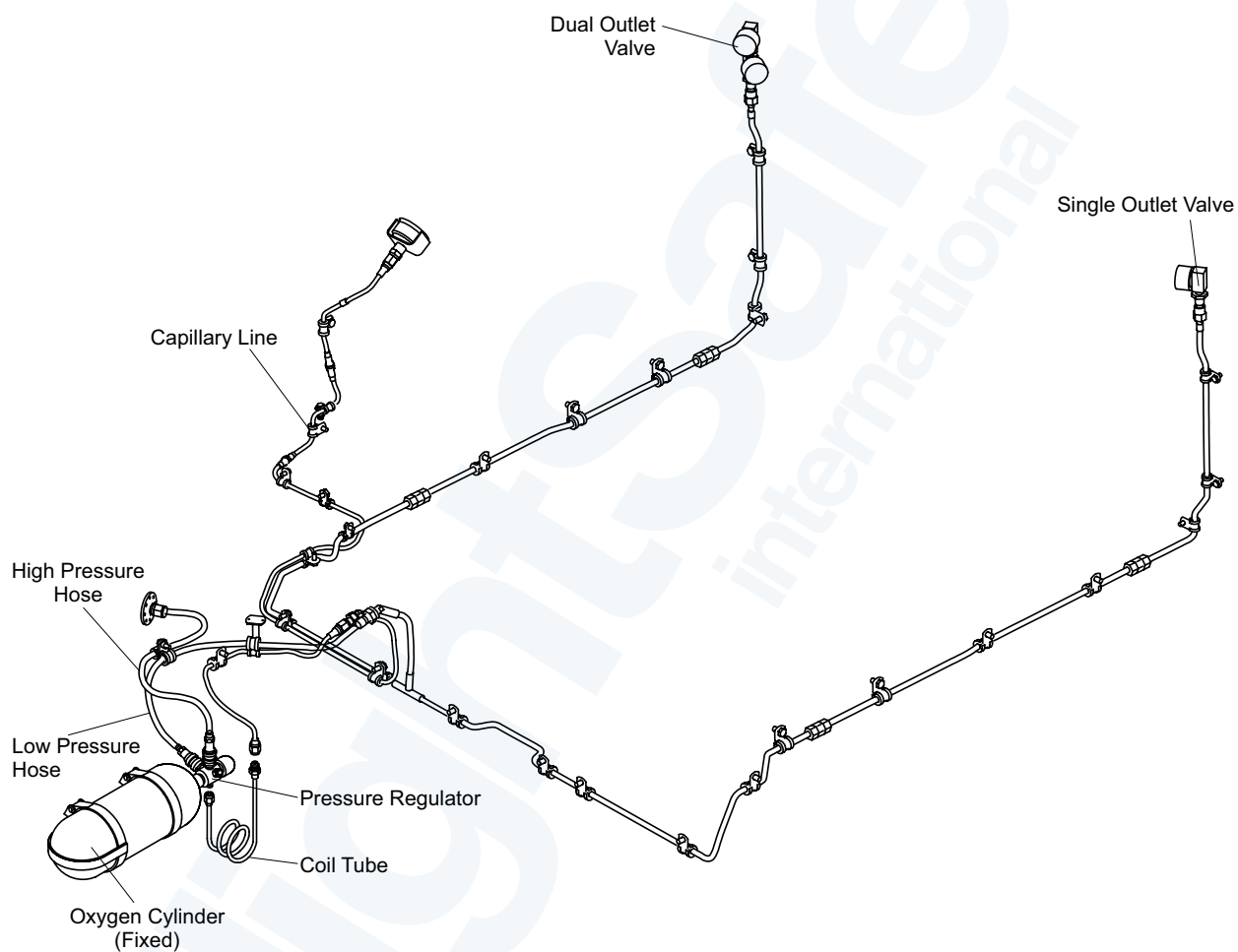
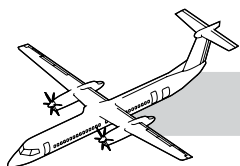
Refer to Figure 35-5. Crew Oxygen System.

A seamless copper alloy tubing capillary line connects the oxygen cylinder to the flight compartment pressure gauge.

Outlet Assemblies

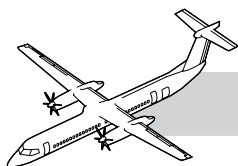
There is a single outlet behind the pilot, and two outlets behind the co-pilot. One of the outlets behind the co-pilot supplies the connection for the observer's mask.

The outlets are equipped with dust covers where they connect with the supply hose of the crew oxygen masks.



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Figure 35-6. Lines and Outlets



Lines and Outlets

Refer to Figure 35-6. Lines and Outlets.

The lines and outlets installed in the flight compartment allow the pilot and the co-pilot to receive oxygen from the cylinder assembly.

The low pressure distribution tubing distributes the oxygen cylinder output pressure of 5-100 psig to the outlet assemblies.

The oxygen outlet valve lets the pilot and co-pilot attach the crew oxygen masks into the crew oxygen system.

Low Pressure Hose Assembly

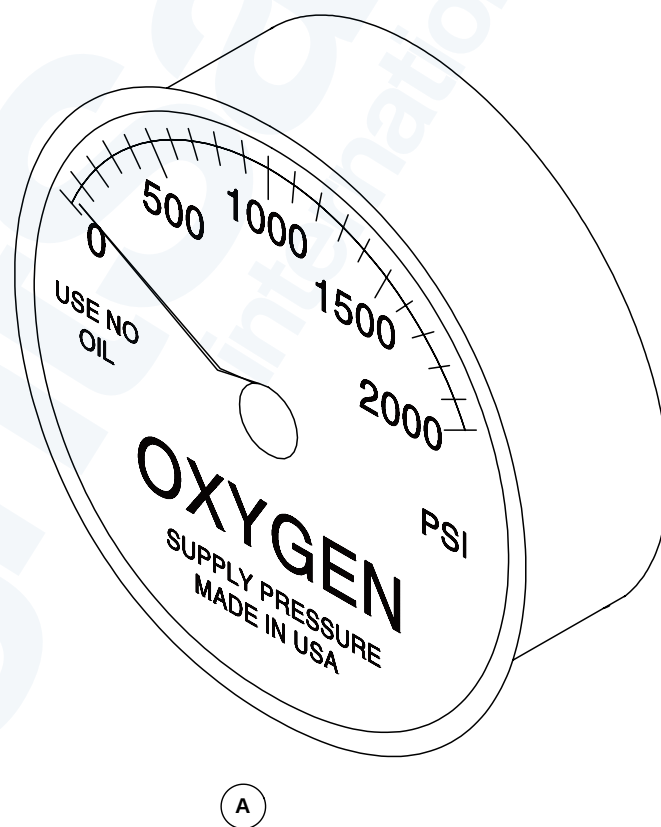
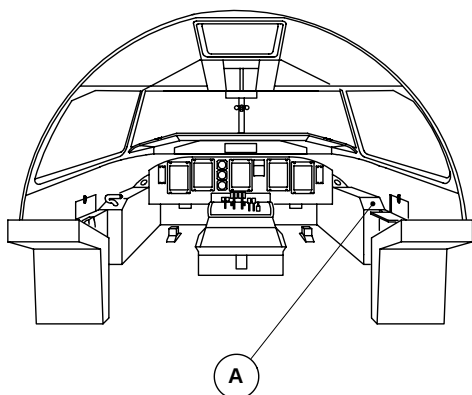
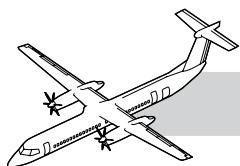
Refer to Figure 35-7. Low Pressure Hose Assembly.

The low pressure hose assembly is attached to the outlet assembly using a quick disconnect. An in-line Oxygen Pressure indicator shows green when normal pressure is sensed and red when the pressure is low.

The hose is made from PVC tubing with nomex (Fire Retarding) braid.

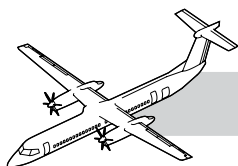


Figure 35-7. Low Pressure Hose Assembly



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Figure 35-8. Flight Compartment Oxygen Pressure Gauge



35-12-00 CREW OXYGEN INDICATION AND SERVICING

If the oxygen cylinder is turned off, the low pressure distribution line pressure is reduced to atmospheric pressure. Residual pressure is trapped in the capillary line and the flight deck indicator will display this trapped pressure.

INTRODUCTION

The Crew Oxygen Indication and Servicing system monitors the crew oxygen system.

GENERAL

This system allows the pilot and co-pilot to accurately find the quantity of oxygen in the system for emergencies.

In the event that the cylinder pressure exceeds 2750 psig it will automatically be released through the Overboard Discharge Indicator.

This system includes the following components:

- Flight Compartment Oxygen Pressure Gauge
- Overboard Discharge Indicator.

COMPONENT DESCRIPTION

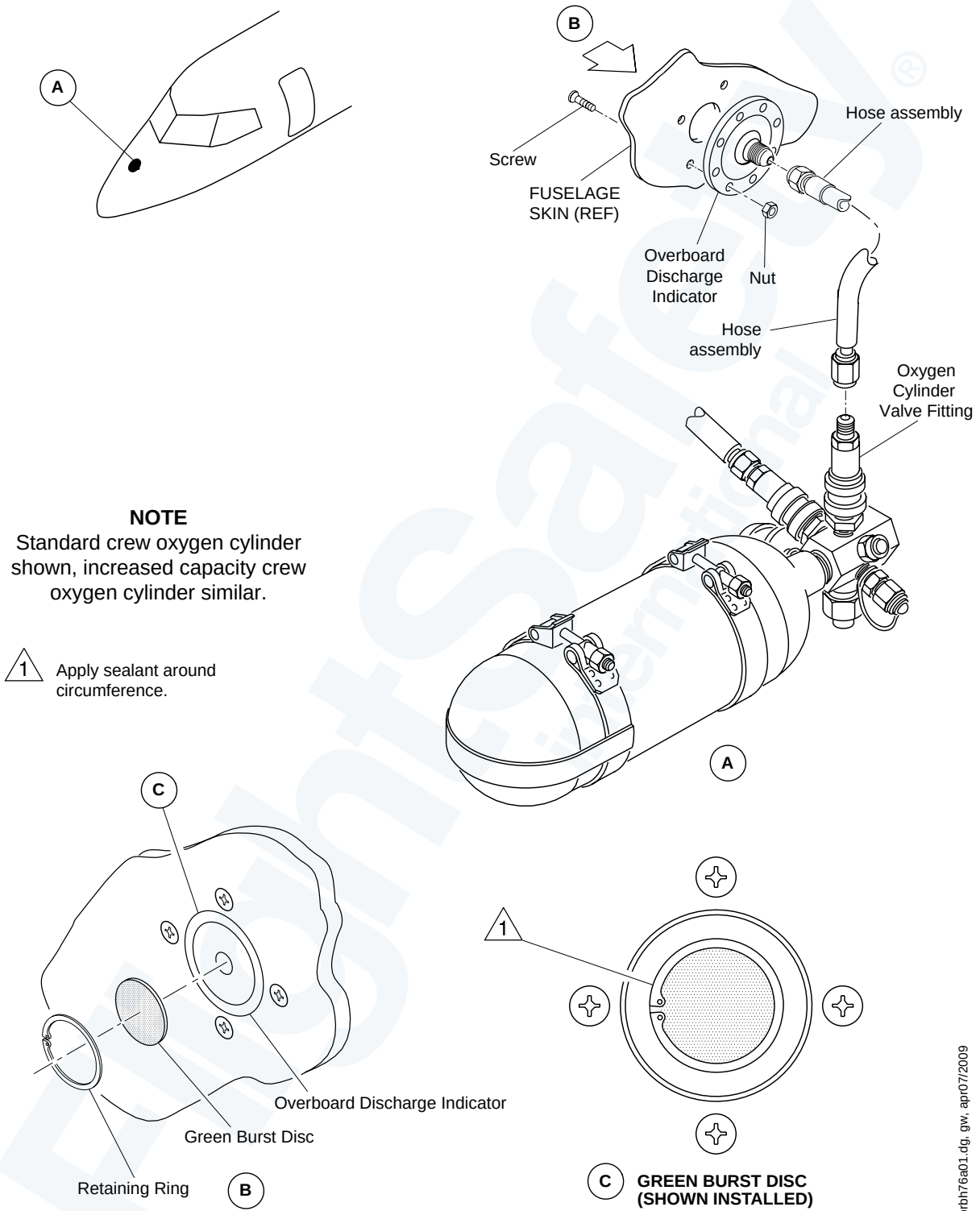
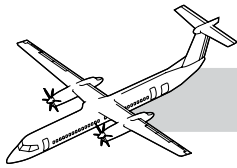
Flight Compartment Oxygen Pressure Gauge

Refer to Figure 35-8. Flight Compartment Oxygen Pressure Gauge.

The flight compartment oxygen pressure gauge, located in the co-pilot side console, shows the amount of oxygen available to the masks.

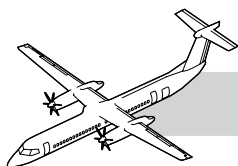
The flight compartment oxygen pressure gauge will supply pressure readings in the range of 0 to 2,000 psig (0 to 13,790 kPag).

It uses a bourdon tube actuator and is burst rated to 3000 psig (20,684 kPag). The gauge has an accuracy of ± 50 psig (345 kPag) over the range at a temperature of 70°F (21.1°C). It has a 5 VDC bulb for lighting. The gauge connects with an adapter to a capillary line which is routed to the cylinder assembly.



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Figure 35-9. Overboard Discharge



Overboard Discharge Indicator

NOTES

Refer to Figure 35-9. Overboard Discharge.

The overboard discharge indicator is located on the nose of the aircraft. The normally green indicator is connected to the high pressure relief port of the oxygen cylinder with flexible hose and quick-disconnect coupling. In the event the oxygen cylinder pressure exceeds 2,750 psig., the excess pressure is released from the cylinder, breaks the green disc in the Overboard Discharge Indicator and discharges excess pressure overboard. The broken/missing disc provides visual indication of overpressure relief of the crew O₂ cylinder.

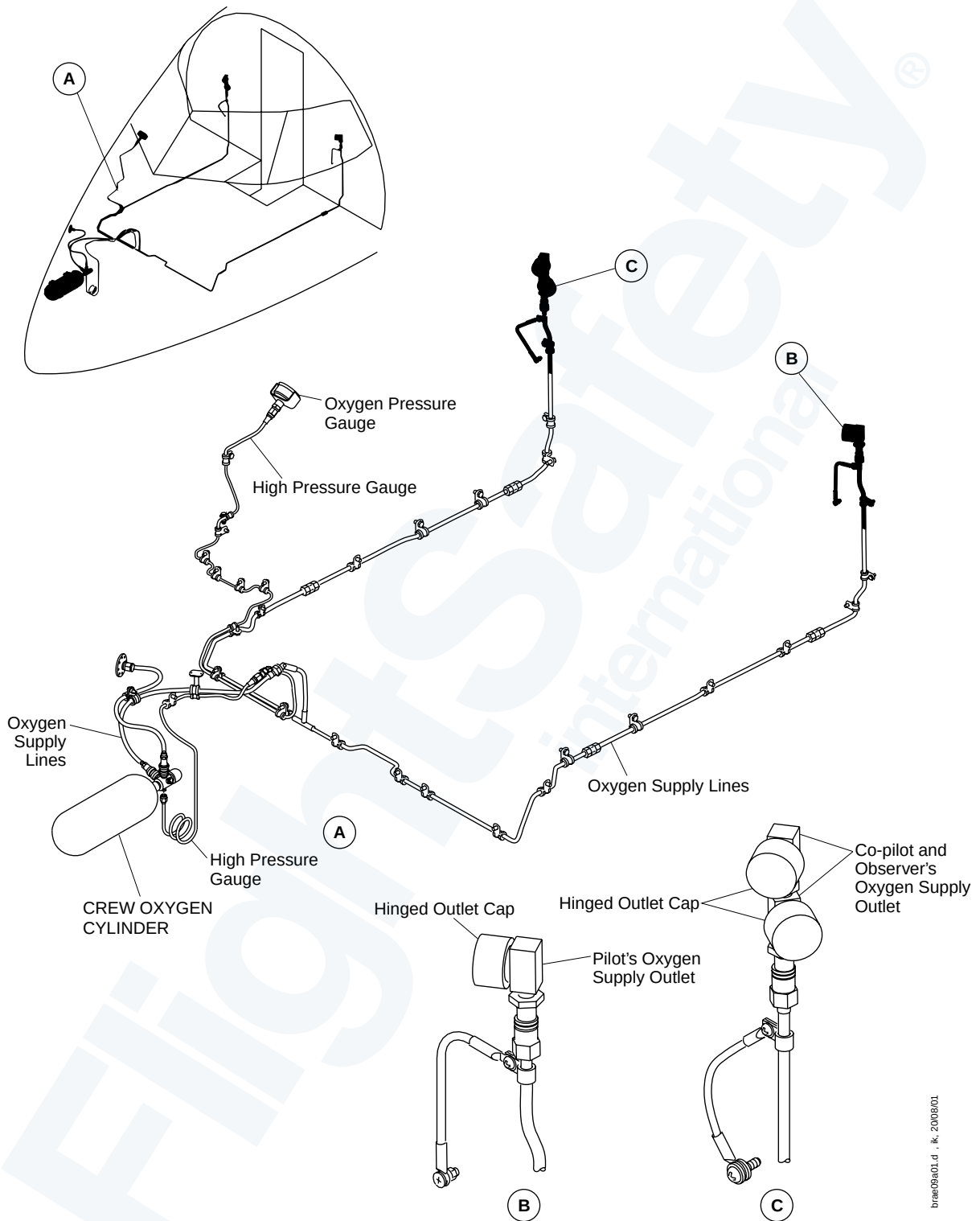
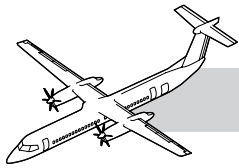
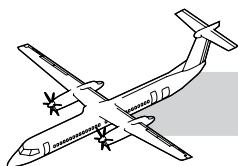


Figure 35-10. Crew Oxygen System - Maintenance Practices



Leak Test of the Crew Oxygen System

NOTES

Refer to Figure 35-10. Crew Oxygen System - Maintenance Practices.

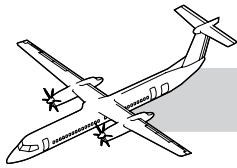
Oxygen system lines must be pressure tested with the high pressure nitrogen test rig to determine if there is system leakage. If there is a leak in the system, necessary repairs are carried out and the test performed again until all leaks are eliminated.

The leak test is done by applying pressure at one end and putting caps on all other open ends. Leaks will be detected by comparing the pressures before and after the testing period. Leaks must be located with applicable leak detector solution.

Bonding test is done before introducing oxygen in the lines to ensure that the conductivity between the oxygen carrying components and the aircraft ground is sufficient to prevent hazardous conditions because of the electrical charges.

Leak test is done in the sequence that follows:

- Pressure test on low pressure lines with nitrogen at 160 psi for 15 minutes
- Purging of low pressure lines with nitrogen at 25 psi pressure for 30 seconds
- Pressure test on high pressure lines with nitrogen at 2,800 psi for 15 minutes
- Purging of high pressure lines with nitrogen at 250 psi pressure for 30 seconds
- Purging of low pressure lines with oxygen at 70 psi. pressure for 15 seconds
- Purging of high pressure lines with oxygen by increasing pressure to 1,850 psi in 10 seconds
- Final leak test with oxygen.



MAINTENANCE PRACTICES

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Flush the Crew Oxygen System

The maintenance procedure that follows is to flush the crew oxygen system.

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO WORK ON OR NEAR THE OXYGEN SYSTEM AND ITS COMPONENTS. IF YOU DO NOT DO THIS, YOU CAN CAUSE A FIRE OR AN EXPLOSION.

Flush the oxygen supply lines (1) with nitrogen as follows:

- Disconnect lines to be flushed and connect nitrogen supply to one end.
- Start the nitrogen flow. Then flush the lines at 25 psi for 30 seconds. Stop the nitrogen flow.
- Disconnect the nitrogen supply.

Flush the oxygen supply lines (1) with oxygen as follows:

- Ensure lines to be flushed are disconnected.
- Connect nitrogen supply to one end and start the oxygen flow.
- Flush the lines at 25 psi for 30 seconds.
- Stop the oxygen flow.
- Disconnect the oxygen supply.

Special Tools

CONNECTOR - OXYGEN OUTLET (Q400)

Part Number: GSB3500001

Purpose

To connect outside nitrogen and oxygen source to the aircraft low pressure oxygen system in order to purge it.

Description

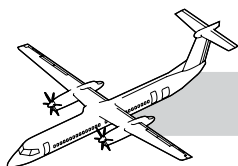
Quick Connector Type

Refer to Figure 35-11. Connector - Oxygen Outlet.



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Figure 35-11. Connector - Oxygen Outlet



Oxygen Servicing Safety Precautions

Obey all the general safety precautions that follow:

- Oxygen Servicing is to be performed by personnel who are familiar with safety precautions.
- Make sure that there is no grease or oil on all clothing, hands, tools, fittings, and oxygen system components.
- Do not smoke in the aircraft or around the aircraft.
- Keep all sources of ignition away from the aircraft.
- When you do work on the oxygen system, park the aircraft in an area with good air flow.
- Put NO SMOKING signs in the applicable areas. Make sure that the signs are in view of other personnel.
- Pressurize or release pressure from all oxygen cylinders in an area with good air flow.
- If bottle pressure is less than 25 psi for longer than 2 hours, purging is required to remove contamination.
- Point the outlet of the oxygen cylinder away from personnel when you release the cylinder pressure.
- Make sure you release the cylinder pressure slowly.
- Do not increase the cylinder pressure at more than 500 psi per minute.
- Make sure that the correct cylinder pressure shows on the cylinder gauge.

Oxygen Safety Precautions

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO WORK ON OR NEAR THE OXYGEN SYSTEM AND ITS COMPONENTS. IF YOU DO NOT DO THIS, YOU CAN CAUSE A FIRE OR AN EXPLOSION.

WARNING

DO NOT USE OIL OR OTHER PETROLEUM LUBRICANTS ON THE OXYGEN SYSTEM COMPONENTS. IF YOU DO THIS, YOU CAN CAUSE A FIRE.

Obey all the general safety precautions that follow:

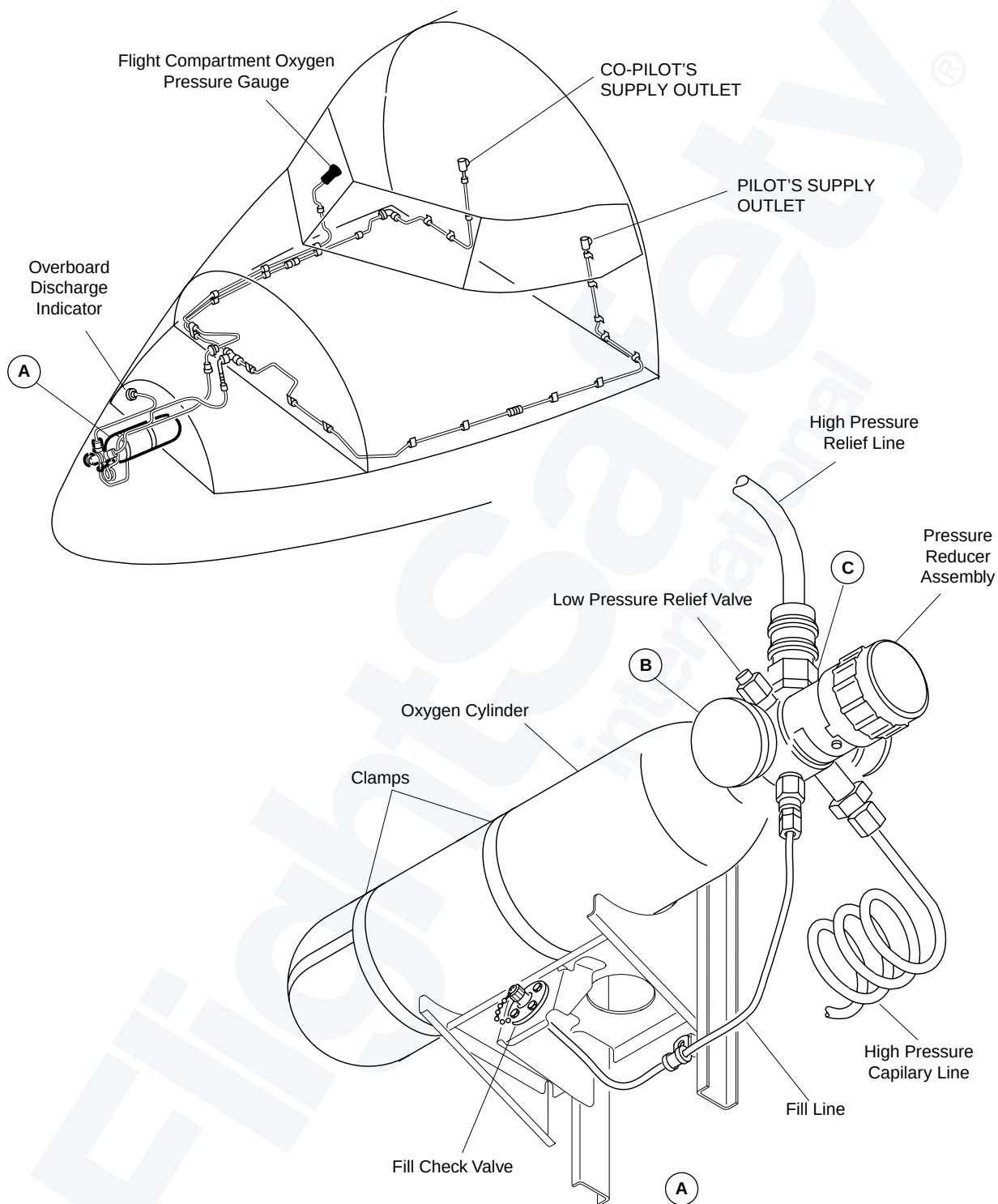
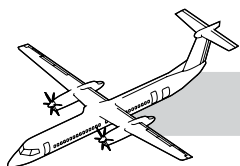
- Ensure work area is clean and well ventilated.
- Remove electrical power from the aircraft.
- Make sure the aircraft is grounded.

Obey all general maintenance precautions that follow:

- Before you remove oxygen components, select the oxygen cylinder to OFF and depressurize the distribution network using the O2 Mask.
- When the maintenance is completed, set the ON/OFF switch on the oxygen cylinder regulator to ON.

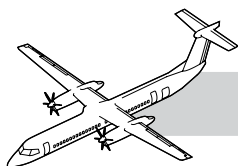
Obey all the precautions that follow:

- Do not use materials other than oxygen leak detection solution on oxygen fittings.
- Use Teflon thread sealing tape only for the threads on the oxygen system pipes.



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Figure 35-12. Servicing of the Crew Cylinder (50 cu. ft. Cylinder)



Servicing of the Crew Oxygen System - 50 cu. ft. Cylinder

Refer to:

- Figure 35-12. Servicing of the Crew Cylinder (50 cu. ft. Cylinder)
- Figure 35-13. Cylinder Pressure/Temperature Correction Graph

The maintenance procedure that follows is for the servicing of the oxygen system.

Pressurize the crew oxygen cylinder as follows:

- Remove the valve cap from the fill check valve (10).
- Attach the hose of the oxygen cart to the fill check valve (10) of the crew oxygen cylinder.
- Monitor temperature that is indicated on the surface thermometer and refer to Figure 35-13 to find the oxygen pressure necessary to fully charge the cylinder.

NOTE

The minimum pressure for dispatch of the aircraft that incorporates the increased capacity crew oxygen cylinder is 1,450 psi for three crew members and 1,050 psi for two crew members at an ambient temperature of 70°F (21.1°C).

- Fully and slowly pressurize the cylinder with the oxygen. Do not increase the pressure more than 500 psi (3,448 kPa) for each minute.
- Stop the oxygen cart when the cylinder gauge shows the full charge pressure per the Pressure/Temperature Correction Graph.
- Slowly remove the hose of the oxygen cart from the crew oxygen cylinder (1). Install the valve cap on the fill check valve (10).

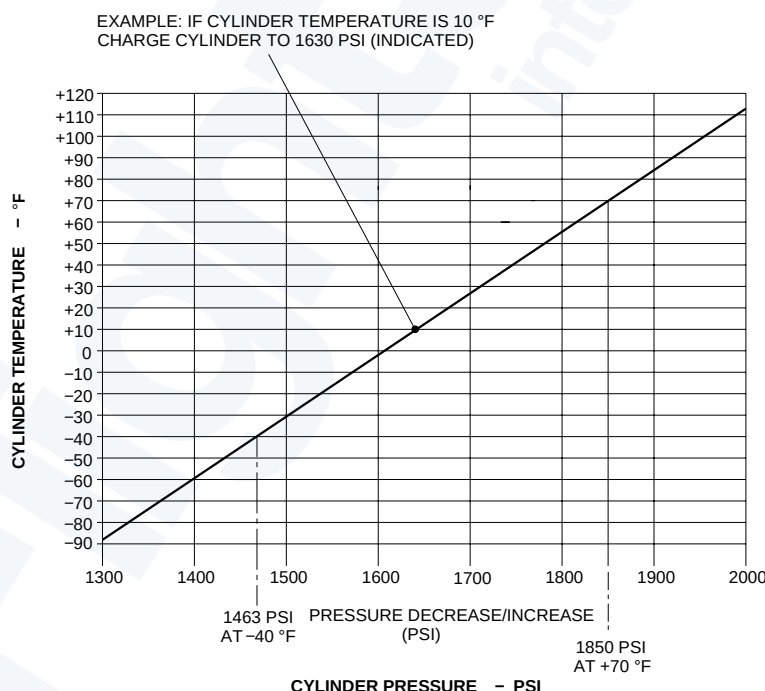
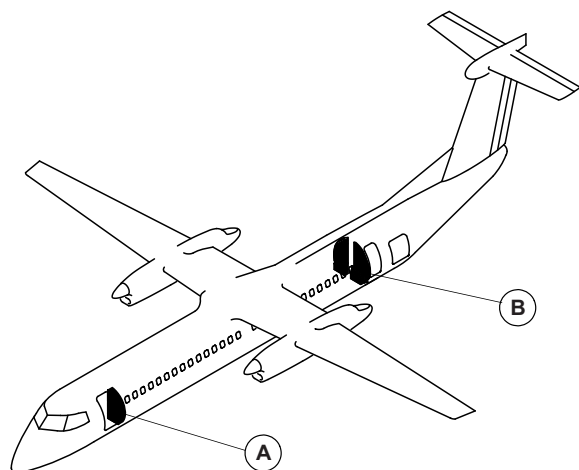
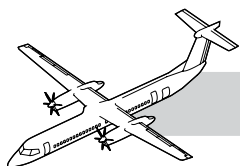
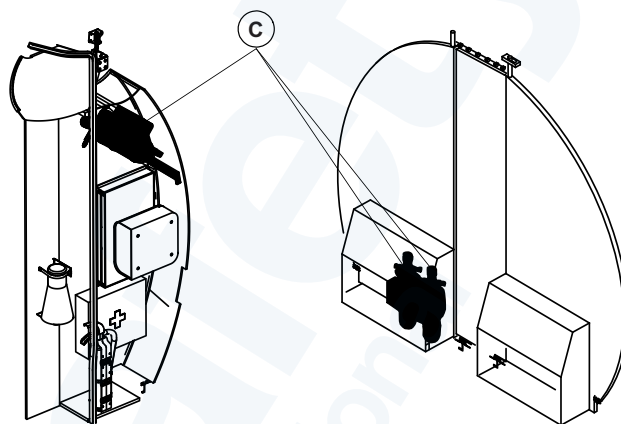


Figure 35-13. Cylinder Pressure/Temperature Correction Graph



NOTE

One container shown.
Other three containers similar.



A FORWARD DRAFT BULKHEAD

B AFT DRAFT BULKHEAD

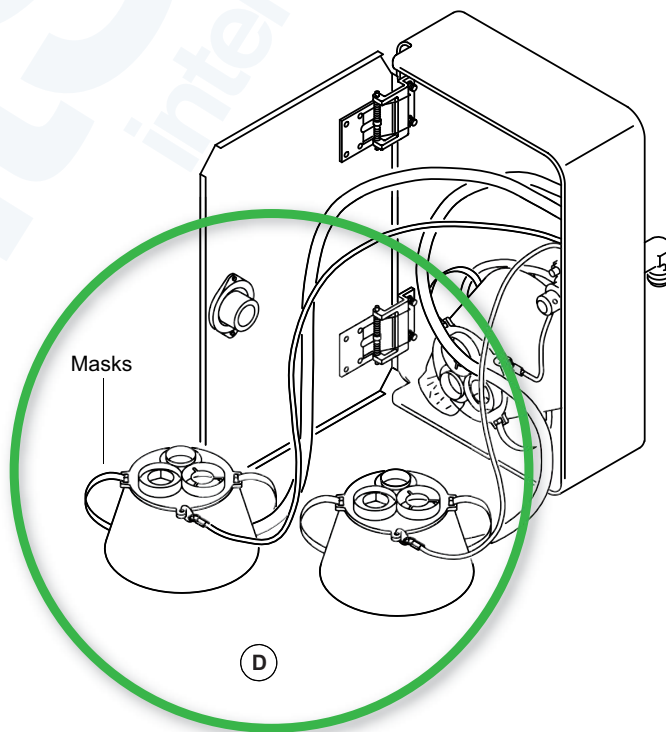
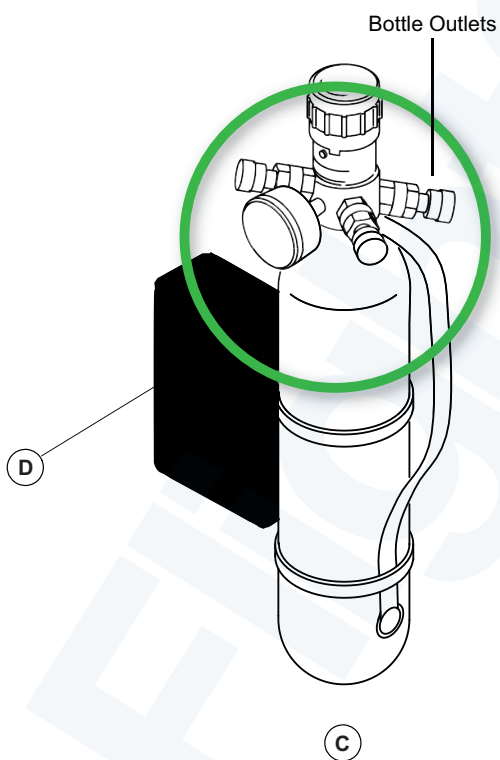
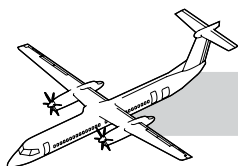


Figure 35-14. Passenger Oxygen System



35-20-00 PASSENGER OXYGEN SYSTEM

NOTES

GENERAL

The passenger oxygen system supplies oxygen for the passengers' use in the event of aircraft depressurization. If the cabin altitude is above 8,000 feet (2,438 meters), the flight attendants can supply oxygen masks to 10 percent of the passengers.

The passenger oxygen system uses three high pressure cylinders with three masks on each cylinder.

SYSTEM DESCRIPTION

Refer to Figure 35-14. Passenger Oxygen System.

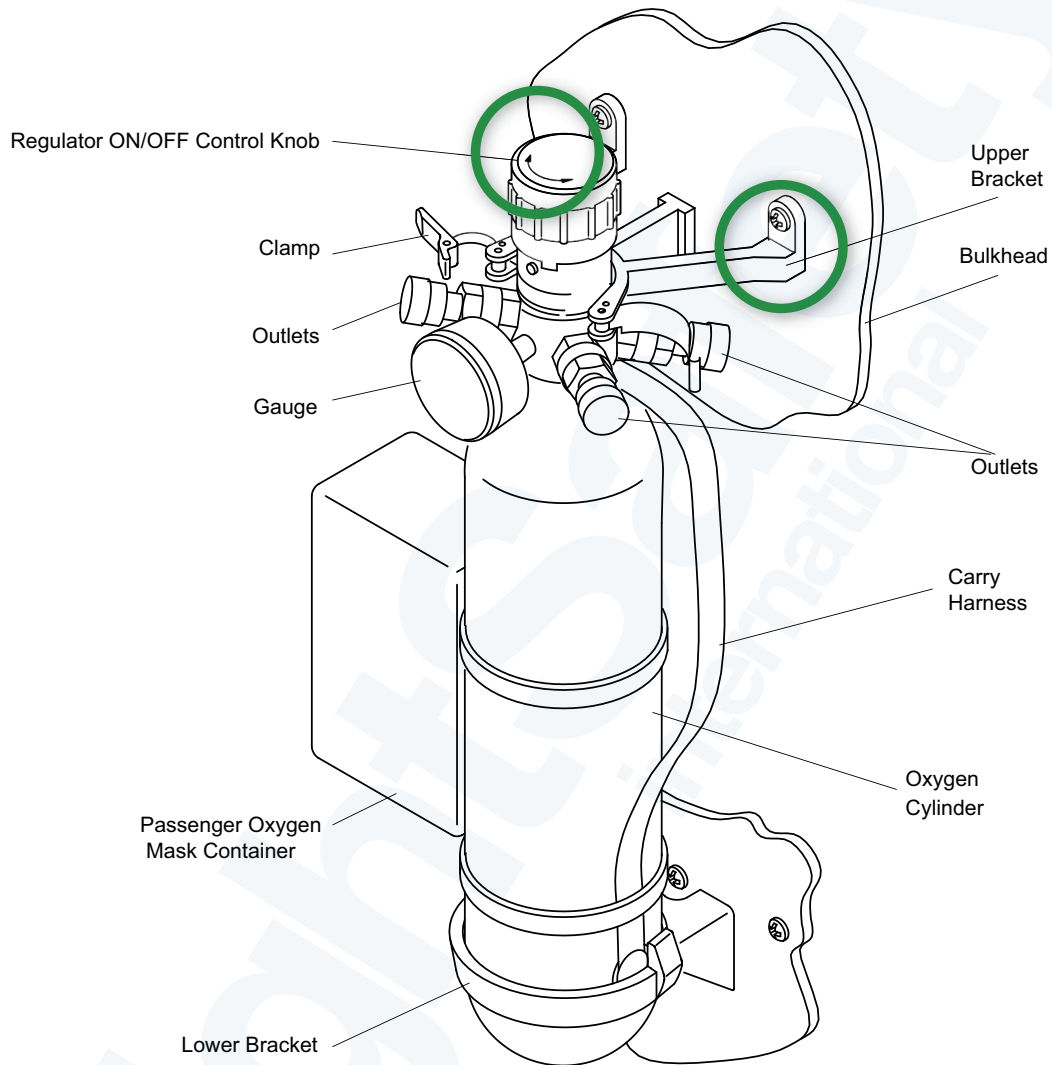
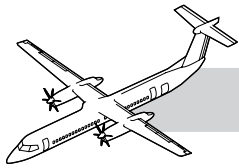
A strap attached to each cylinder, is used to support the cylinder by the passenger or aircraft seat.

The oxygen mask hose line is connected to the bottle outlet and the mask is adjusted to fit the face of the passenger.

Operating the green control knob to the FULL ON position starts the oxygen flow which is indicated by a flow indicator and the expansion of the mask breathing bag.

The green control knob is turned to the FULL OFF position to stop the oxygen flow.

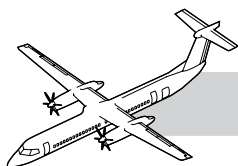
To refill, the cylinder is removed from the aircraft and serviced to the rated pressure via the filler adapter.



NOTE

One passenger oxygen system shown.
Other two systems are similar.

Figure 35-15. Passenger Oxygen Cylinders



COMPONENT DESCRIPTION

High Pressure Cylinders

Refer to Figure 35-15. Passenger Oxygen Cylinders.

The cylinder assemblies are made up of a composite construction and are serviced to 1,800 psig to provide 11 ft³ capacity. The maximum operational pressure is 5,500 psig. and the minimum dispatch pressure is 1,720 psig at 70° F (21.1° C).

Pressure Regulators

The pressure regulator of each cylinder assembly has a fail safe design. The cylinder assembly has a fill port, three mask outlets, pressure gauge and a safety fitting with an integral burst disk. The regulator reduces cylinder pressure to 70 psig (483 kPag) for delivery to the masks. Each regulator has an integral low pressure relief valve, to prevent false indication of mask/bottle pressure when the control knob is turned off.

A frangible burst disc is installed on the regulator body to provide venting of the bottle if the pressure reaches 2,495 to 2,775 psig at 60°F (16.5°C).

The outlets are calibrated to let a maximum of 2.0 L/min flow at 25,000 ft (7,620m) altitude.

Masks

Refer to Figure 35-14. Passenger Oxygen System.

Each oxygen cylinder supplies three polyvinyl masks connected by silicone tubing.

Upper and Lower Retaining Brackets

Refer to Figure 35-15. Passenger Oxygen Cylinders.

The brackets support the cylinder assemblies in stowed position. The upper bracket has a quick release mechanism.

Operational Test of the Passenger Oxygen System

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO WORK ON OR NEAR THE OXYGEN SYSTEM AND ITS COMPONENTS. IF YOU DO NOT DO THIS, YOU CAN CAUSE A FIRE OR AN EXPLOSION.

Make sure the portable oxygen cylinder is charged to 1800 psi.

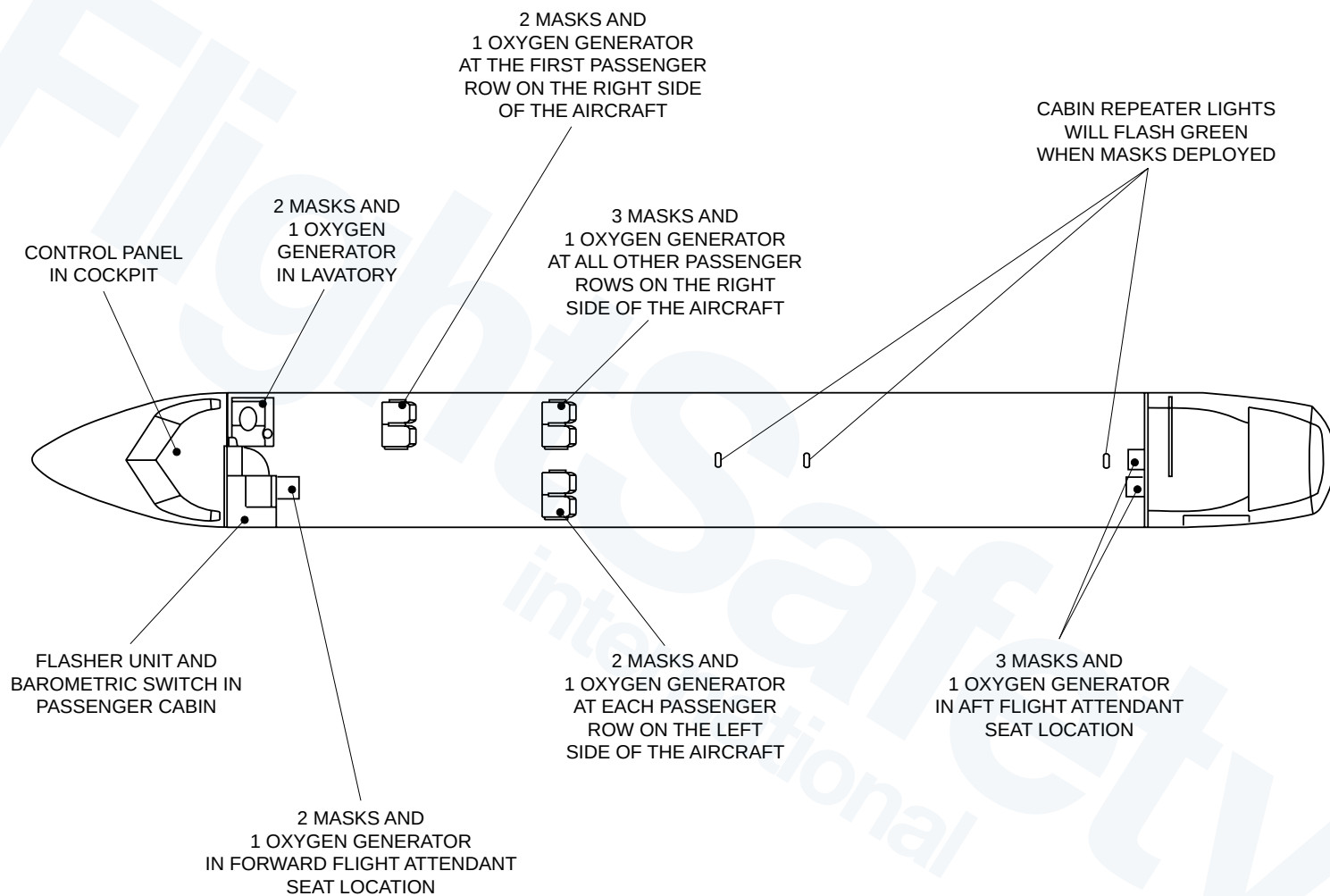
Do the operational test of the passenger oxygen system with each mask as follows:

- Remove the masks from the storage bag
- Make sure that each mask is connected to an outlet on the portable oxygen Cylinder.
- Turn the regulator ON/OFF control on the portable oxygen cylinder to the ON position.
- Breathe through each mask to make sure there is flow of oxygen to the mask.

NOTE

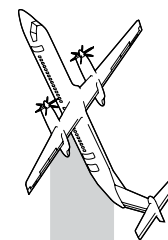
The flow indicator in the line to the mask should be green to indicate the flow of oxygen.

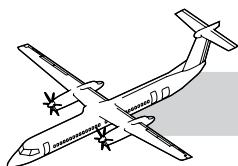
- Turn the regulator ON/OFF control on the portable oxygen cylinder to the OFF.
- Breathe through the mask until you can breathe easily.
- Make sure the flow indicator in the line to each mask is red.



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Figure 35-16. Drop-down Oxygen System





35-21-00 PASSENGER DROP-DOWN OXYGEN SYSTEM

NOTES

INTRODUCTION

Refer to:

- Figure 35-16. Drop-down Oxygen System.
- Figure 35-17. Drop-Down Oxygen Assembly - Forward and Aft Lavatories.
- Figure 35-18. Set-up for Passenger Oxygen Drop-Down System Test.

On aircraft with SB84-25-98 or ModSum 4-458569 and 4-458665 incorporated, the passenger drop-down oxygen system is used to supply oxygen to the passengers if the aircraft cabin depressurizes.

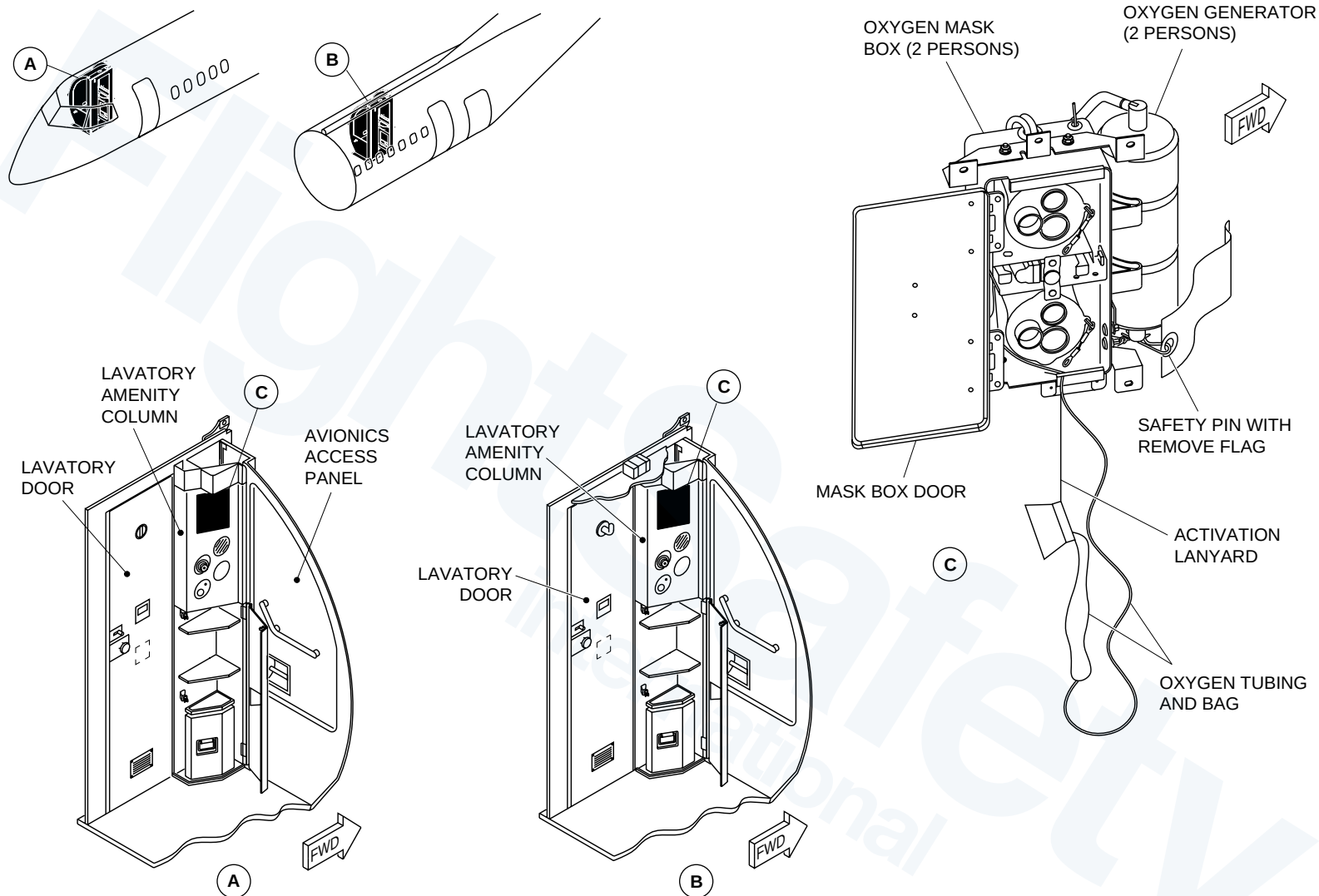
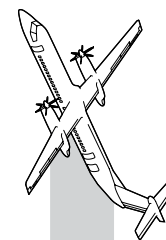


Figure 35-17. Drop-Down Oxygen Assembly - Forward and Aft Lavatories



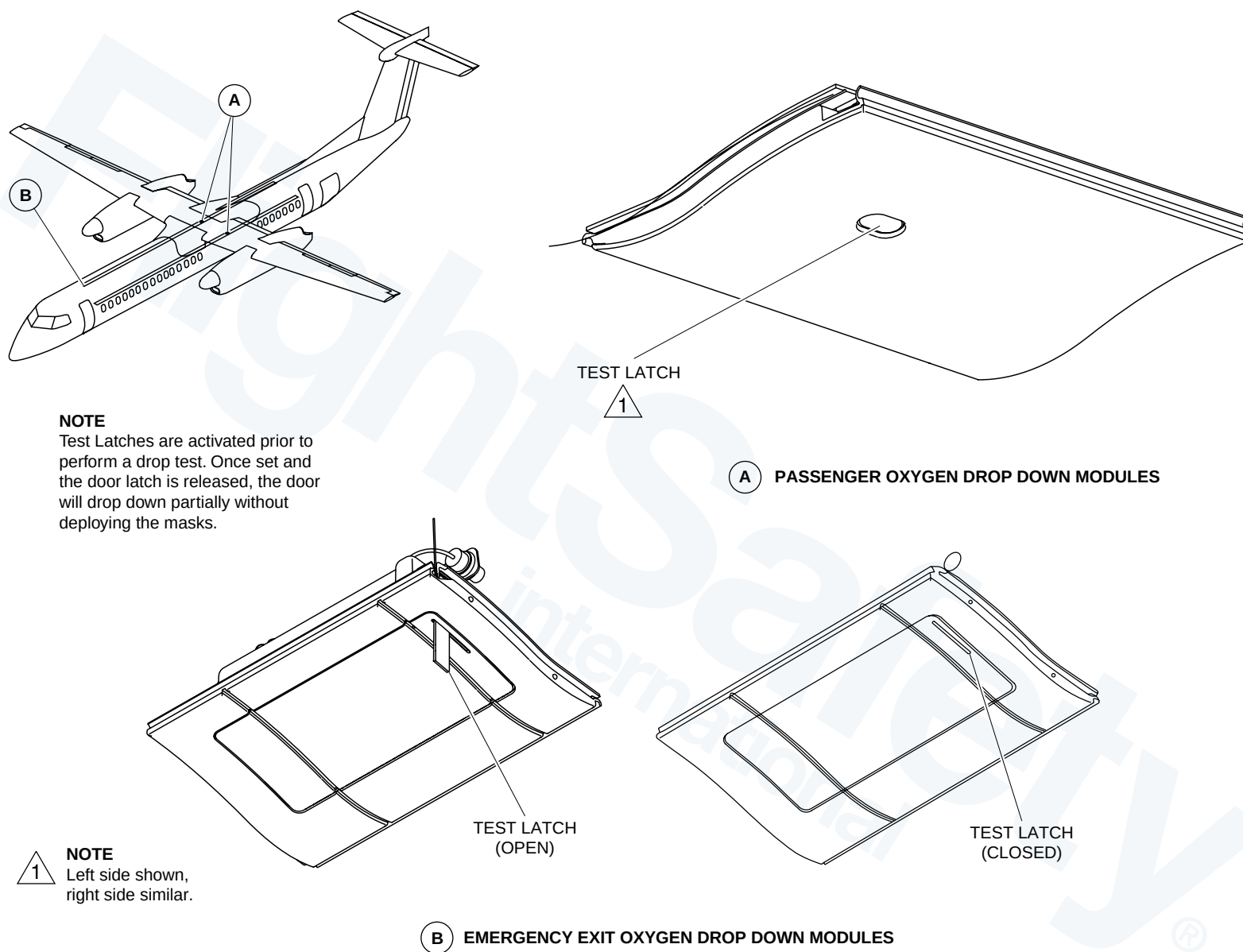
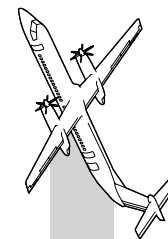


Figure 35-18. Set-up for Passenger Oxygen Drop-Down System Test

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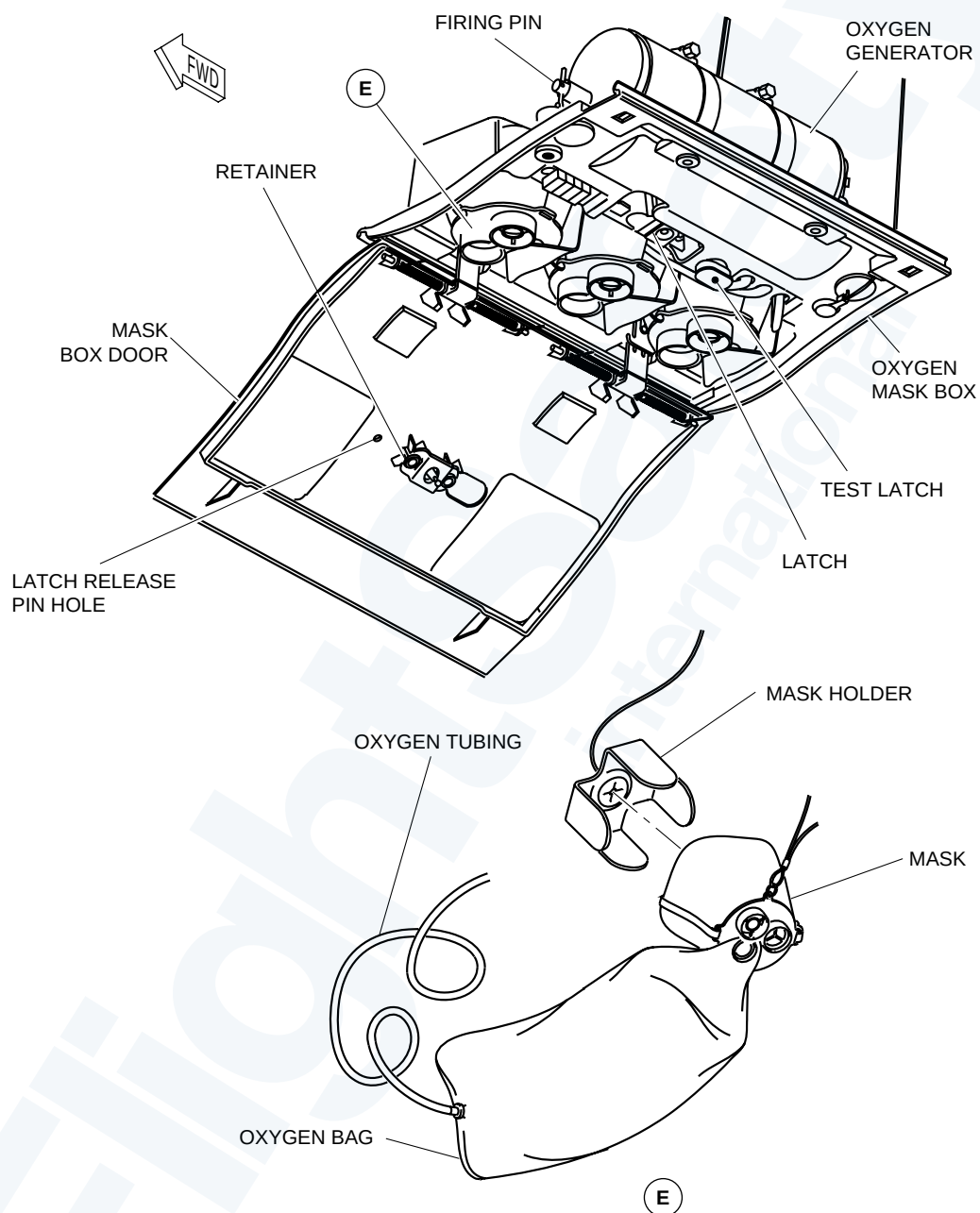
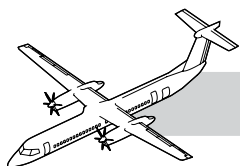
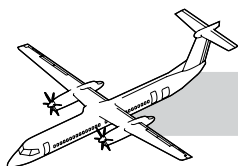


Figure 35-19. Drop-Down Oxygen Assembly - Passenger Compartment



COMPONENT DESCRIPTION

NOTES

Oxygen Mask Box

Refer to Figure 35-19. Drop-Down Oxygen Assembly - Passenger Compartment.

The oxygen mask boxes are installed above each pair of passenger seats, each flight attendant station, and in the forward lavatory.

Refer to Figure 35-19 for location of Lavatory installations.

The oxygen mask box includes electrically controlled, mechanical latch used to release the mask box doors, allowing the contained oxygen masks to drop. There is a provision to operate the latch with a manual deploy tool.

Test Latches are incorporated in the mask box door mechanism. These latches are set by maintenance prior to testing the deployment of the drop-down oxygen system. They allow partial door opening that will confirm the electro-mechanical latch release has taken place without dropping the oxygen mask.

Refer to Figure 35-20 for the different types of Test Latches.

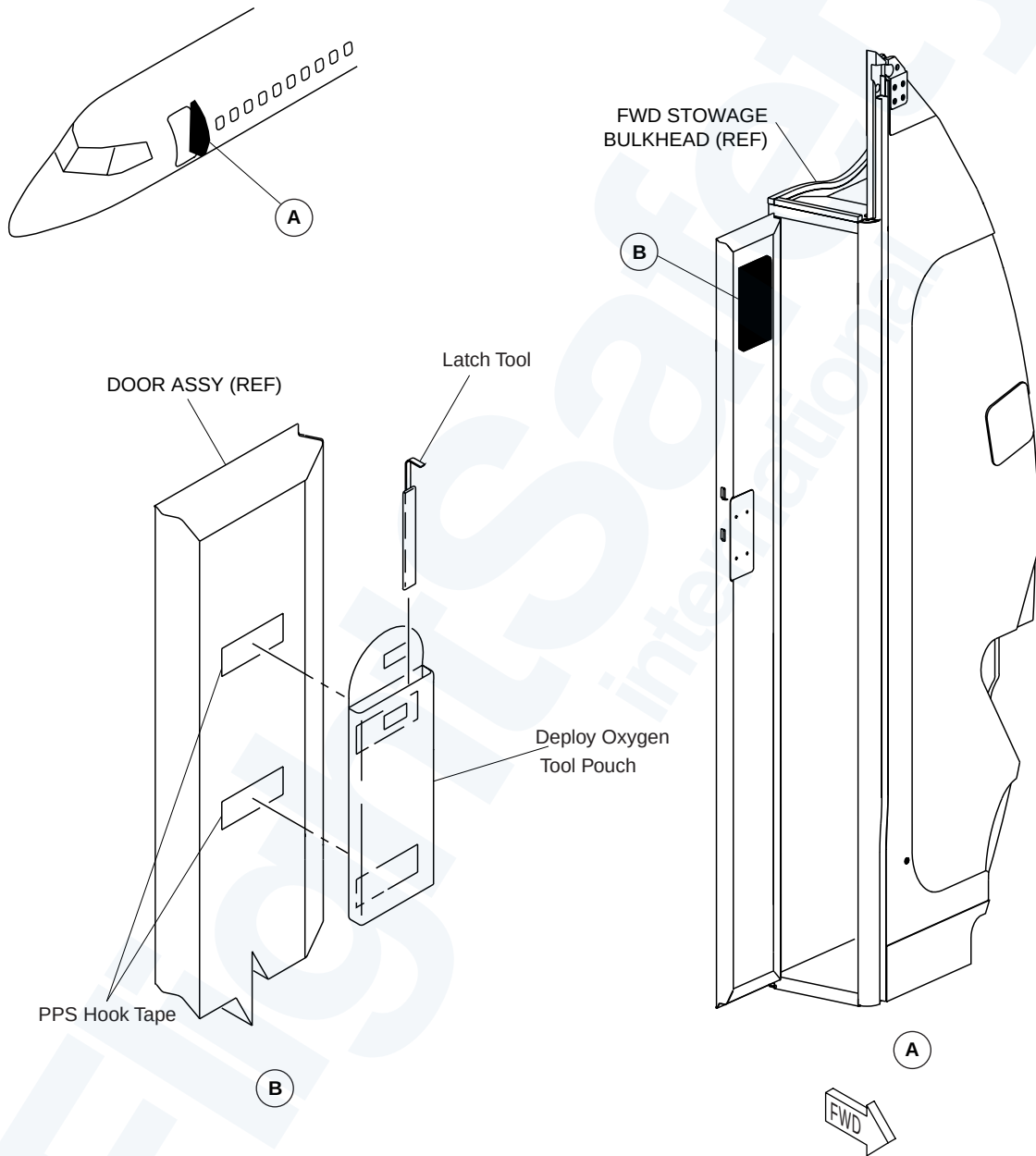
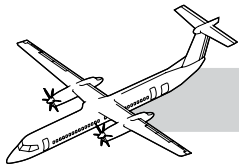
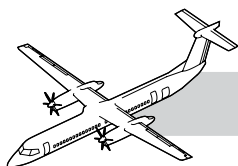


Figure 35-20. Manual Deploy Oxygen Tool



Manual Deploy Tool

NOTES

Refer to Figure 35-20. Manual Deploy Oxygen Tool.

One manual deploy tool is kept in the forward stowage bulkhead in a cloth pouch. The pouch is attached on the inside of the forward stowage bulkhead door.

The straight end of the tool is used to open a mask box door that failed to open during mask deployment.

The hook end of the tool is used to pull out the 'test latches' on the drop-down oxygen assemblies before a system test.

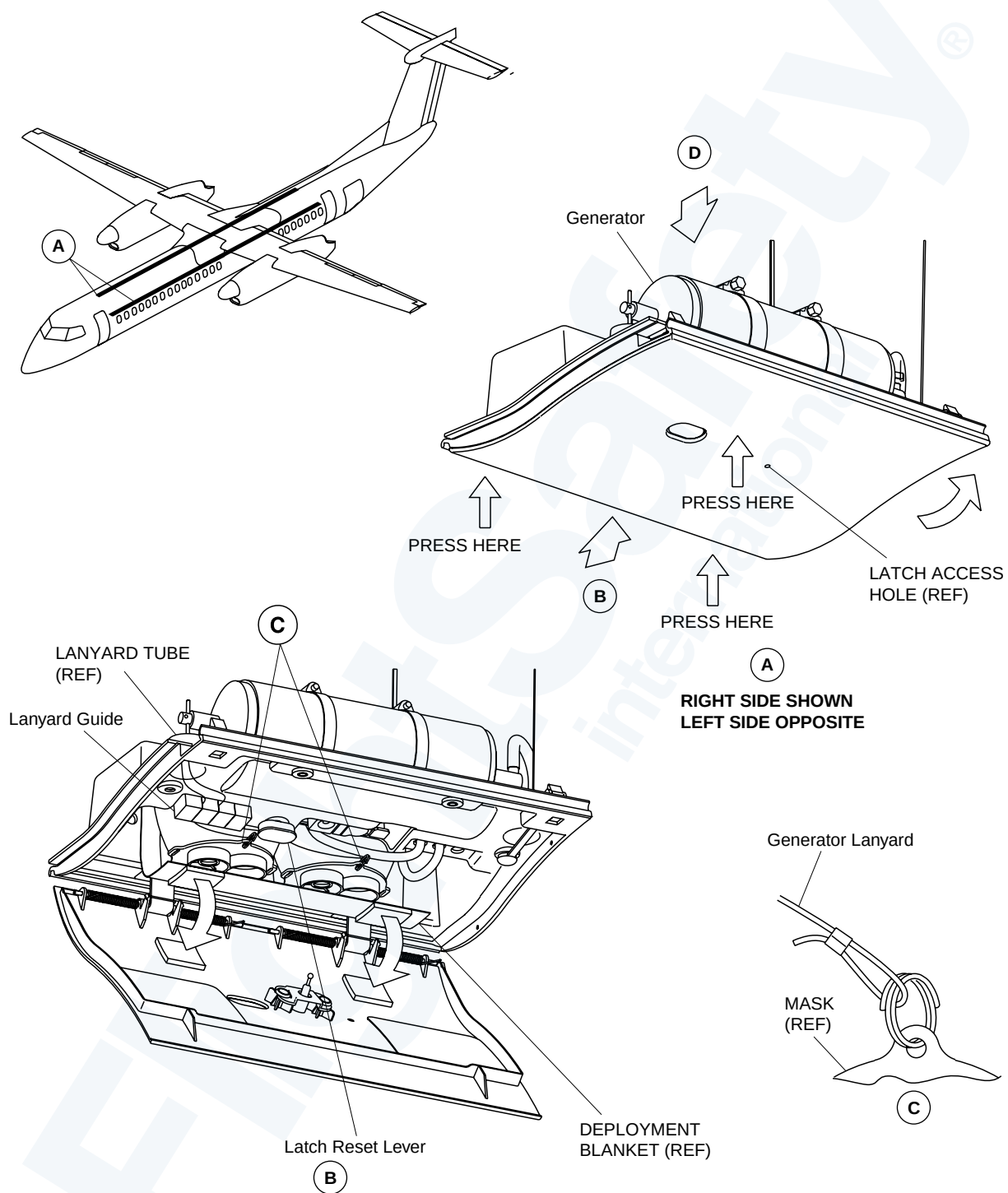
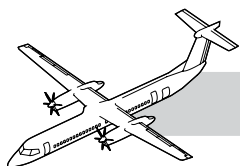
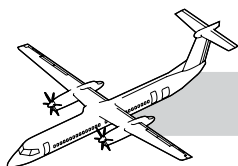


Figure 35-21. PSU Oxygen Generator



Oxygen Generator

Refer to Figure 35-21. PSU Oxygen Generator.

The generators supply emergency oxygen to the passenger masks when the system is deployed and the masks are used.

Each chemical oxygen generator has a steel cylinder, a firing pin, a release pin, and an output port. The steel cylinder holds the chemical contents that are used to make oxygen.

The release pin is located on the firing pin and is connected to the lanyard.

When a force applied to the lanyard of 5-6 lbs. is applied the firing pin is released and the chemical reaction inside the cylinder produces gaseous oxygen for a maximum of 22 minutes. A yellow indicator on the generator case, below the firing pin, turns black due to the high heat produced by the chemical reaction when the generator has been activated.

The oxygen generators are line replaceable units.

Latch Mechanism

The latch mechanisms are solenoid operated and electrically-controlled. When energized, the mechanisms releases the oxygen door latches to be released and allow the masks to drop-down. The latch mechanism can also be released or reset manually for maintenance with the manual deploy tool.

Passenger Mask

The passenger masks are plastic oxygen masks with reservoir air bags. They are connected to the chemical oxygen generators with nylon hoses.

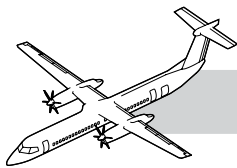
Each passenger mask has a lanyard, a nylon feed tube, elastic straps, a plastic face cup, a spring-loaded flow indicator, a reservoir bag, and three check valves (intake, exhaust, and ambient-air check valve).

When the passengers pull the masks to their faces, the lanyard pulls the release pin on the chemical oxygen generator and activates the chemical generator.

When the flight attendant Oxygen Mask Box door is opened, the flight attendant pulls the yellow 'PULL lanyard' to operate the chemical oxygen generator release pin.

The passenger and flight attendants place the mask plastic face cups on their noses and mouths and position the elastic straps over their head to hold the mask in place.

The oxygen flows through the feed tube, flow indicator and into the reservoir bag. The flow indicator shows 'green' when oxygen flow exists. When the passenger breathes in, the intake valve opens and oxygen flows into the plastic face cup. When the passenger breathes out, the air pushes the exhaust check valve open and the air flows out of the plastic face cup. The valve opens to supply additional air to the passenger if the oxygen flow into the mask is not sufficient.



Passenger Oxygen Control Panel

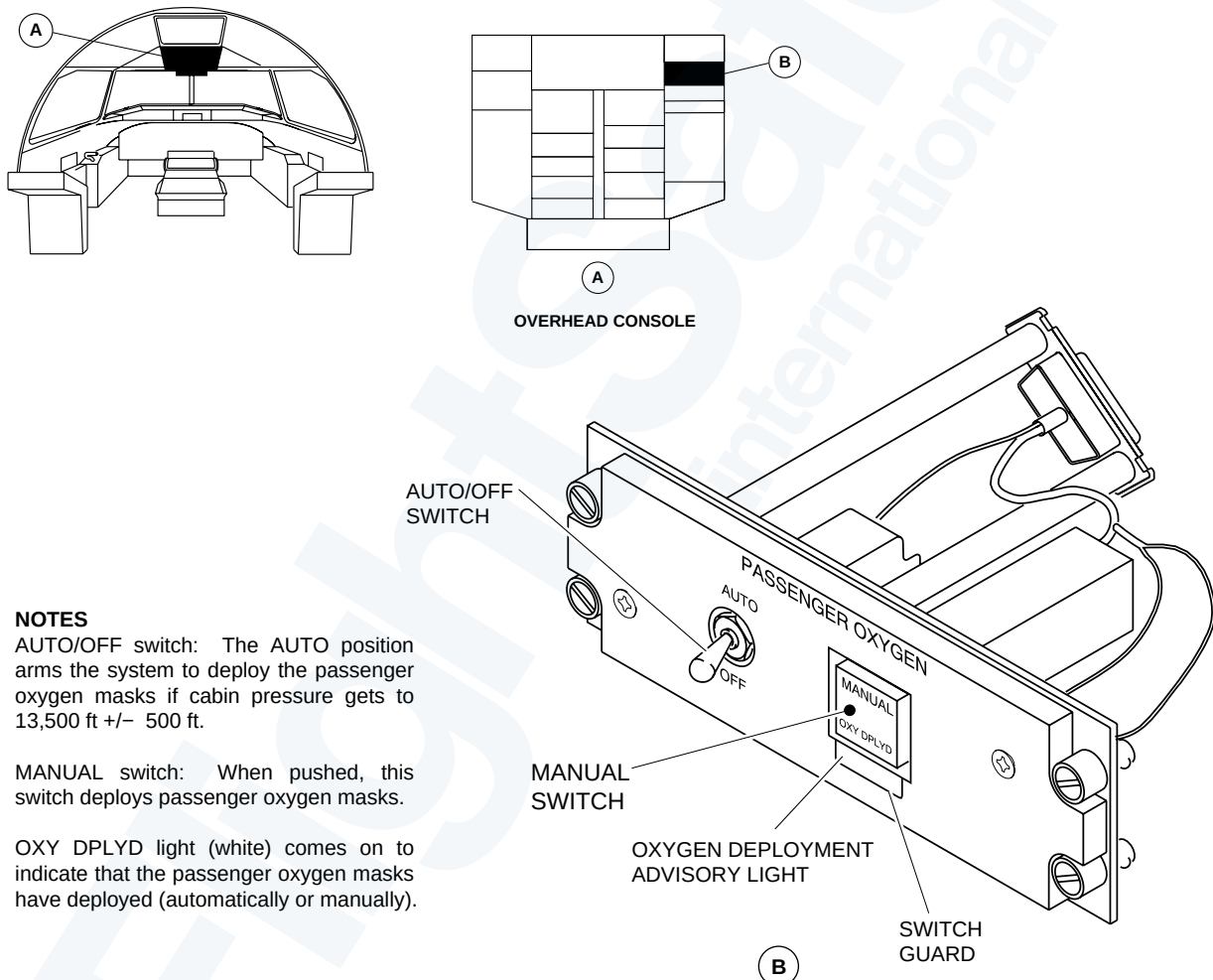
Refer to Figure 35-22. Passenger Oxygen Auto/Manual Switch Facing Panel.

The passenger oxygen control panel is installed on the right side of the overhead panel in the flight compartment. There is an AUTO/OFF toggle switch and a guarded MANUAL pushbutton switch on the passenger oxygen control panel.

A white OXY DPLYD advisory light is in the MANUAL pushbutton switch assembly comes on when the oxygen masks are deployed automatically or manually.

In the AUTO selection, the drop-down oxygen masks will deploy automatically when the barometric pressure switch is activated.

MANUAL switch selection by the flight crew, activates the drop-down oxygen masks immediately.



NOTES

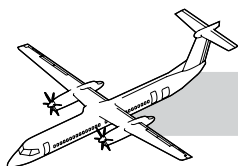
AUTO/OFF switch: The AUTO position arms the system to deploy the passenger oxygen masks if cabin pressure gets to 13,500 ft +/- 500 ft.

MANUAL switch: When pushed, this switch deploys passenger oxygen masks.

OXY DPLYD light (white) comes on to indicate that the passenger oxygen masks have deployed (automatically or manually).

POST SB84-25-98 OR POST MODSUM 4-458569

Figure 35-22. Passenger Oxygen Auto/Manual Switch Facing Panel



Barometric Pressure Switch/ Flasher

Refer to Figure 35-23. Barometric Switch.

Installed on the relay panel for the drop-down oxygen system in the avionics rack.

The Pressure switch supports the automatic deployment of the passenger drop-down oxygen masks when the cabin pressure senses $13,500 \pm 500$ ft.

The Flasher causes the Green oxygen LED on the all three Advisory Light Panels to flash when the drop-down oxygen mask system is deployed automatically or manually.

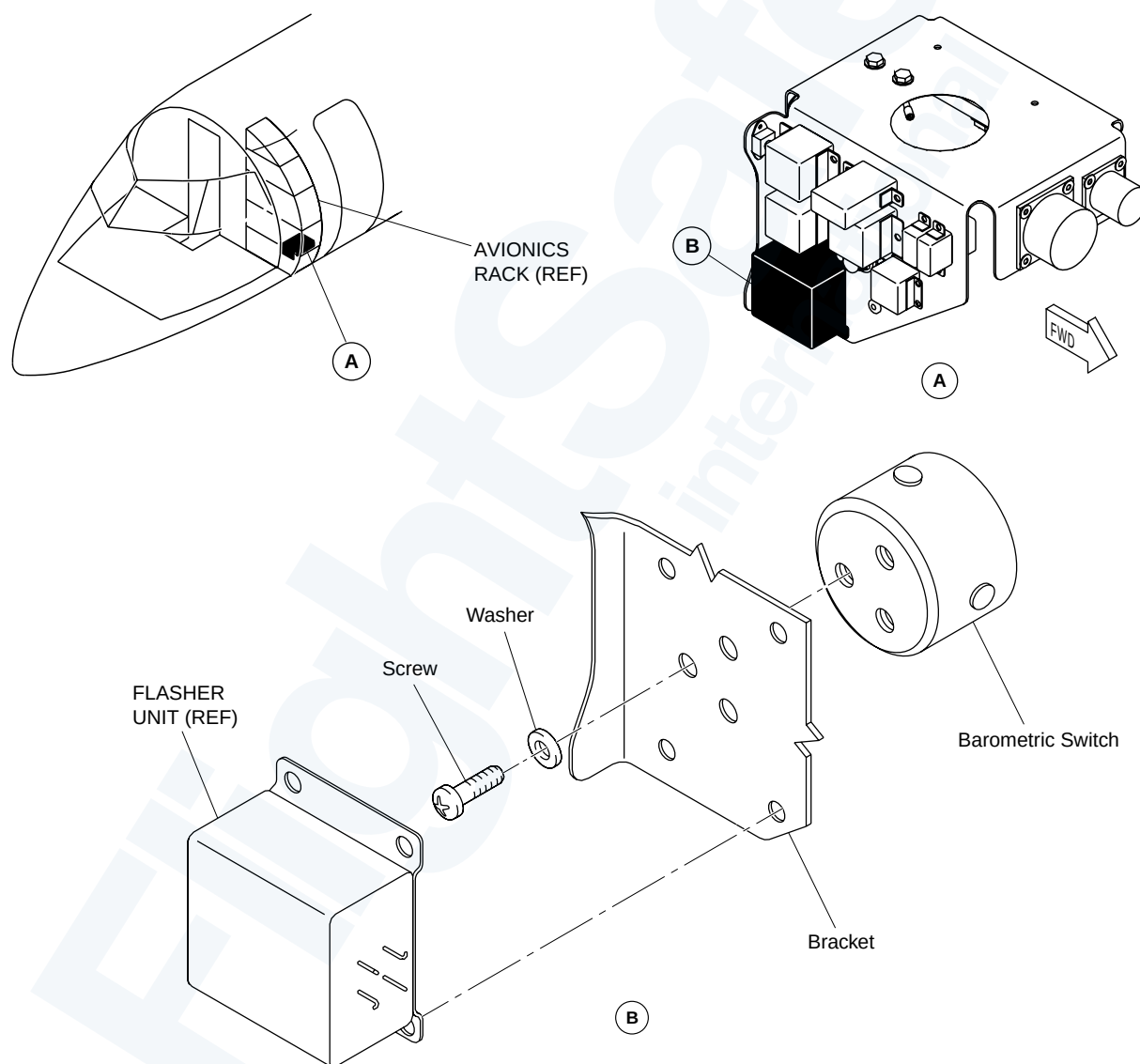


Figure 35-23. Barometric Switch

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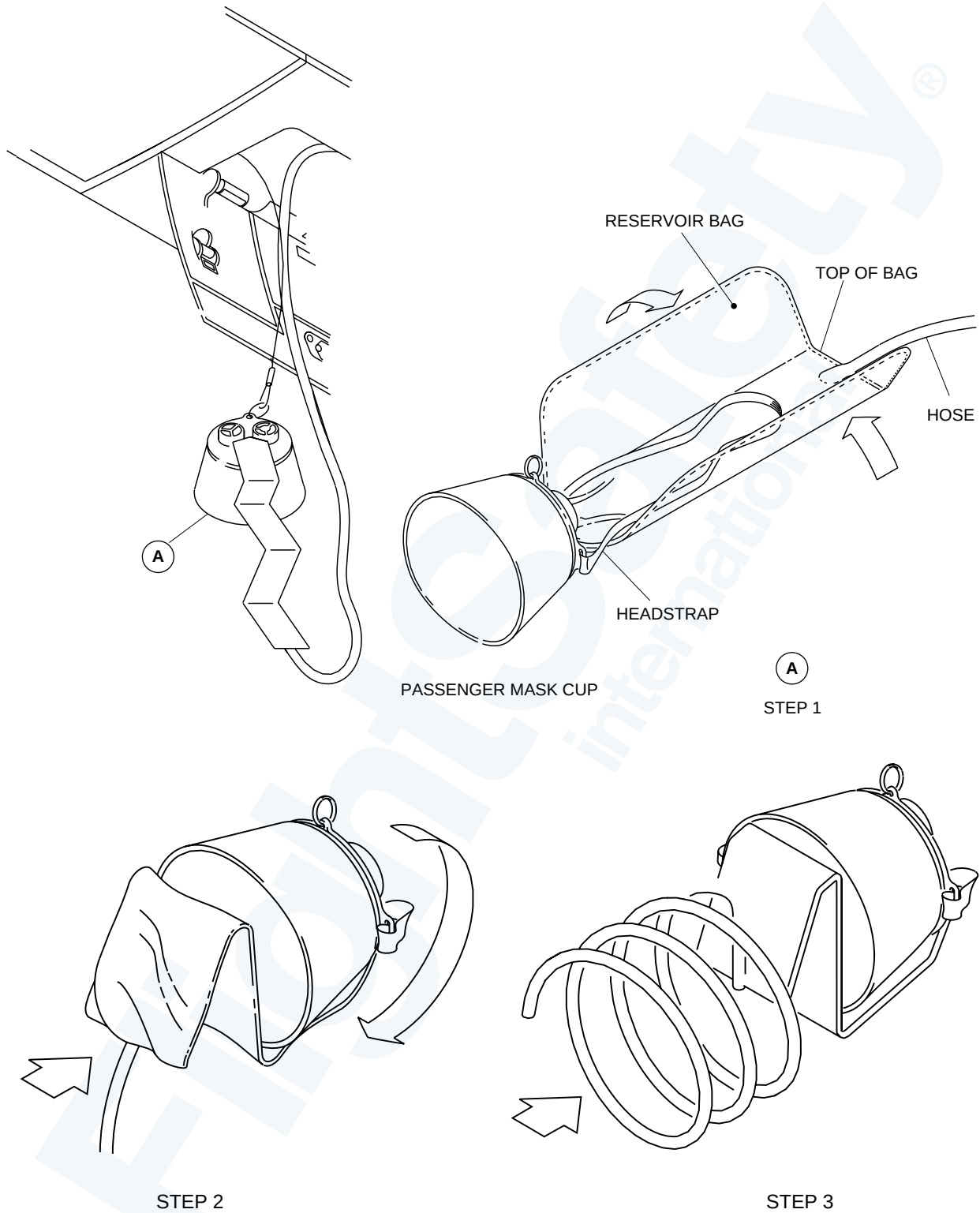
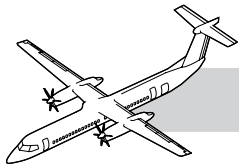
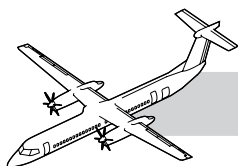


Figure 35-24. Repacking of the Passenger Oxygen Mask (1 of 2)



Repacking of the Passenger Oxygen Mask

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to:

- Figure 35-24. Repacking of the Passenger Oxygen Mask (1 of 2)
- Figure 35-25. Repacking of the Passenger Oxygen Mask (2 of 2)

The maintenance procedure that follows is to repack the passenger oxygen mask assembly.

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO WORK ON OR NEAR THE OXYGEN SYSTEM AND ITS COMPONENTS. IF YOU DO NOT DO THIS, YOU CAN CAUSE A FIRE OR AN EXPLOSION.

Repack the PSU mask after deployment as follows:

- Reset the door latch by pivoting the latch reset lever.
- Make sure that the deployment blanket is properly attached to the door.

With each mask repack, have an assistant hold the mask pack while the subsequent task is performed.

Place each mask up into the mask container and pull the door partially shut.

NOTE

Masks must be placed to prevent the hoses and lanyards from becoming tangled.

- Route each lanyard through the lanyard guide. Store excess lanyard length on the door under each mask.
- Close the door completely by pressing firmly in the middle until the door latches.
- Make sure the door is locked.
- Press on the two hinge side corners of the door and then along the hinge edge of the door to seat the hinges.

Flight Attendant, Lavatory and Emergency Exit repacking procedures are similar.

Subtask 35-20-00-860-002/3/4 and 5 provides instruction that are unique to the installation.

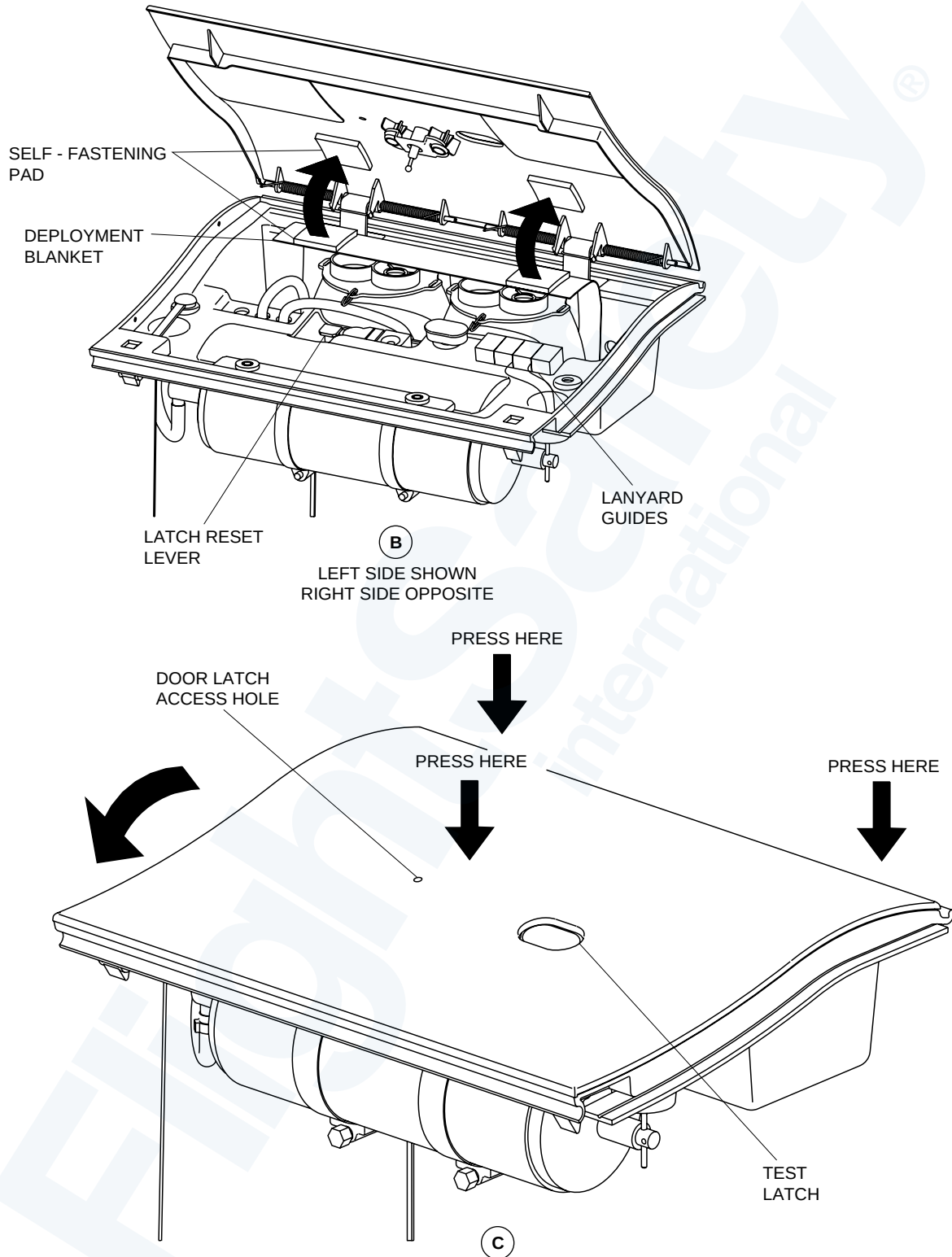
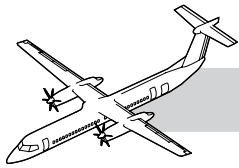
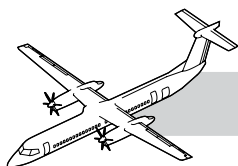
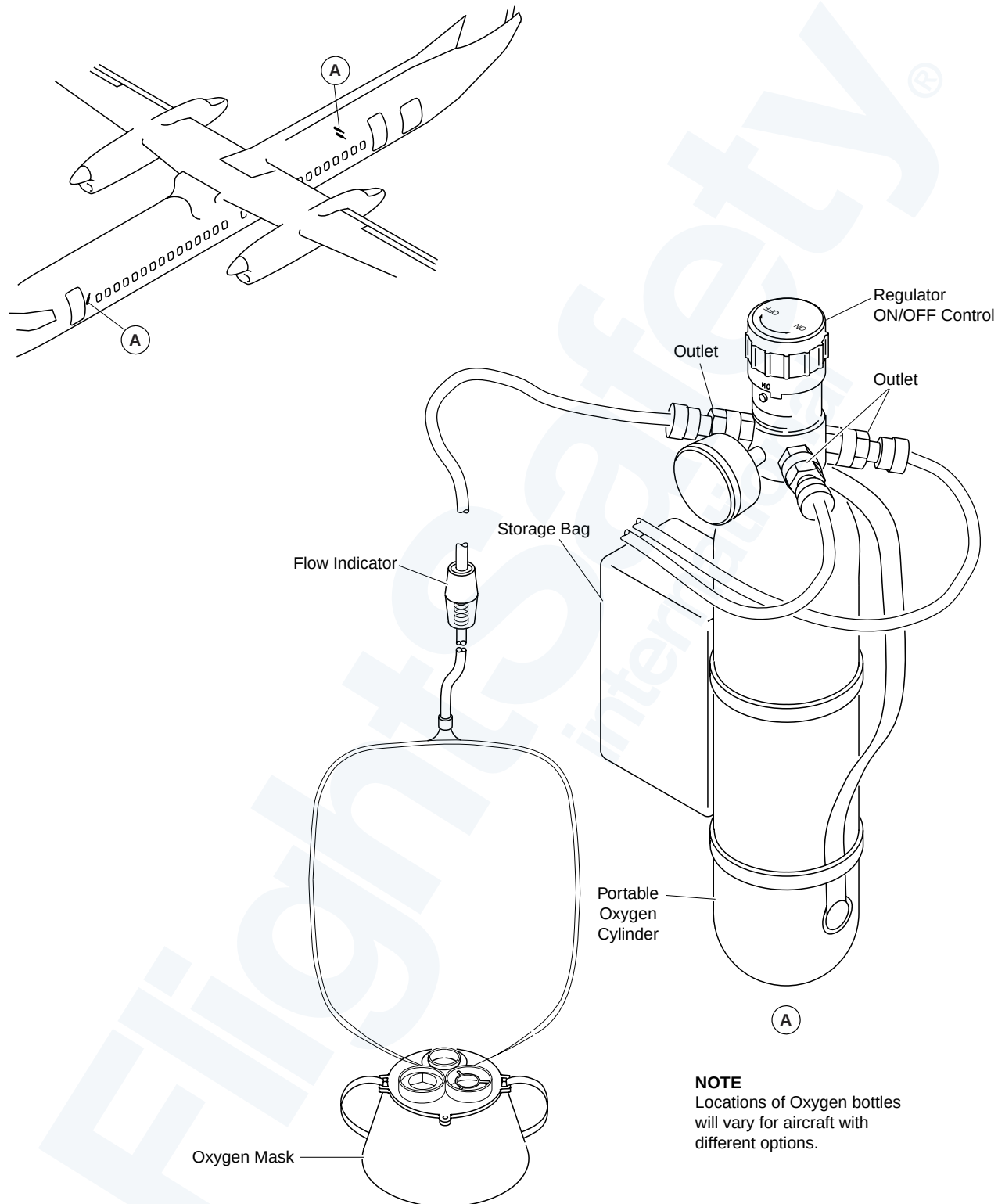
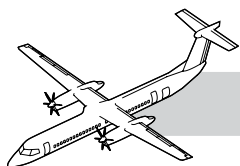


Figure 35-25. Repacking of the Passenger Oxygen Mask (2 of 2)

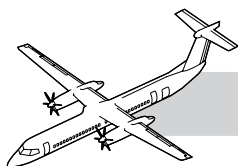


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Figure 35-26. Portable Flight Attendant Oxygen Cylinder Assembly



35-30-00 PORTABLE OXYGEN SYSTEM

NOTES

INTRODUCTION

The crew portable oxygen system has:

- Portable flight attendant oxygen cylinder assembly (optional)
- Protective breathing equipment (PBE) units.

COMPONENT DESCRIPTION

Flight Attendant Oxygen

Refer to Figure 35-26. Portable Flight Attendant Oxygen Cylinder Assembly.

A portable oxygen system, dedicated to the flight attendant, is available as a customer option. The flight attendant's portable oxygen system has:

- A bottle assembly with pressure gauge
- A constant flow outlet
- A low pressure relief valve
- A high pressure frangible safety plug
- An on-off valve
- A charging valve
- A constant flow mask with a flow indicator and a 50 inch (1.27 meter) long connecting hose
- A mask reservoir bag and a carrying strap.

The flight attendant oxygen system is intended for single person use. It has a mask connected by a hose to the outlet of an oxygen bottle.

A placard located on the outside of the galley compartment identifies the contents inside. Another placard located inside the compartment identifies the minimum dispatch pressure and the appropriate service manual references for maximum charge pressure and inspection intervals.

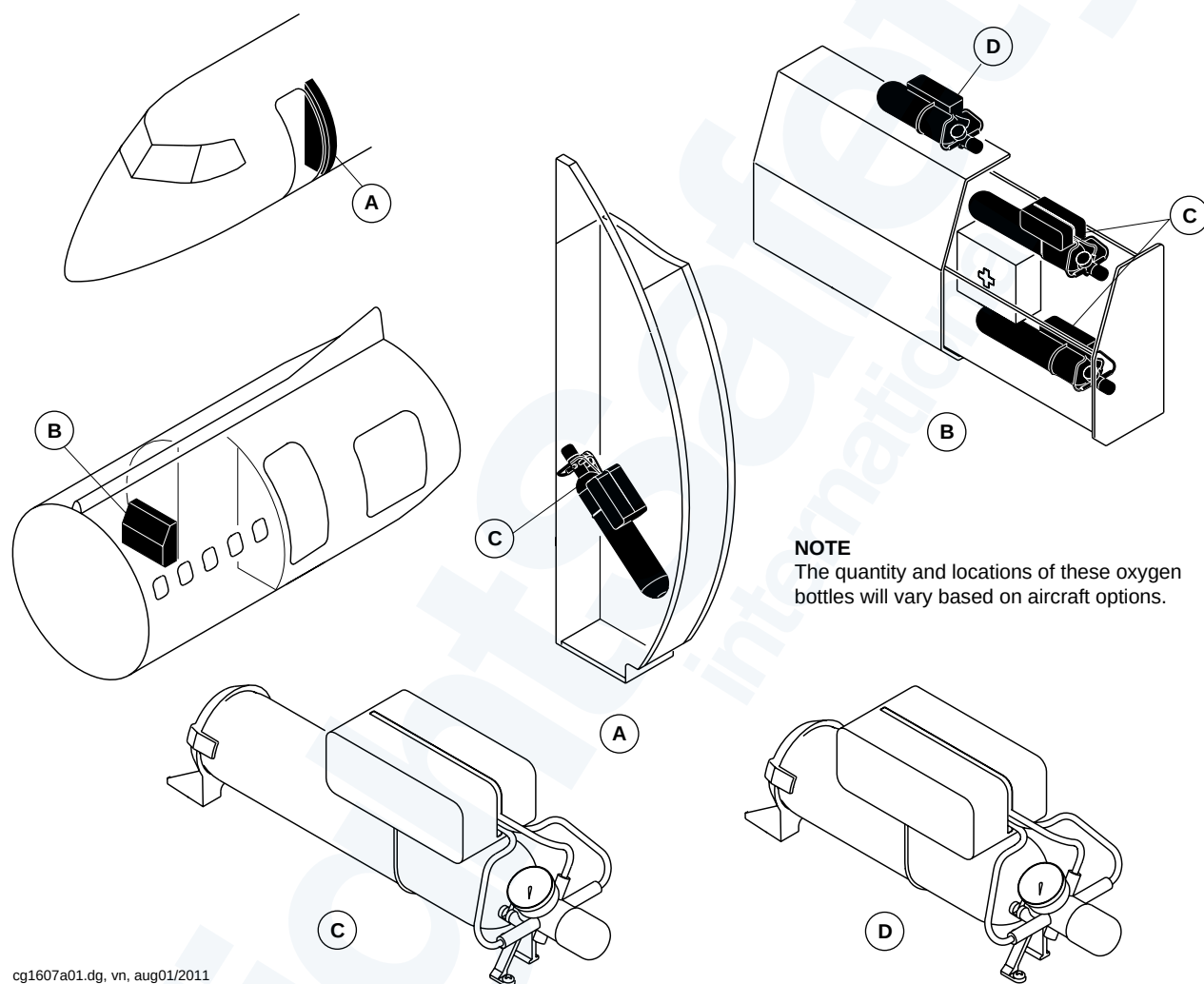
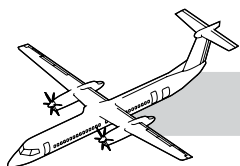
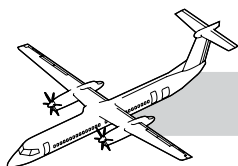


Figure 35-27. Passenger Emergency Oxygen System



Passenger Emergency Oxygen System

NOTES

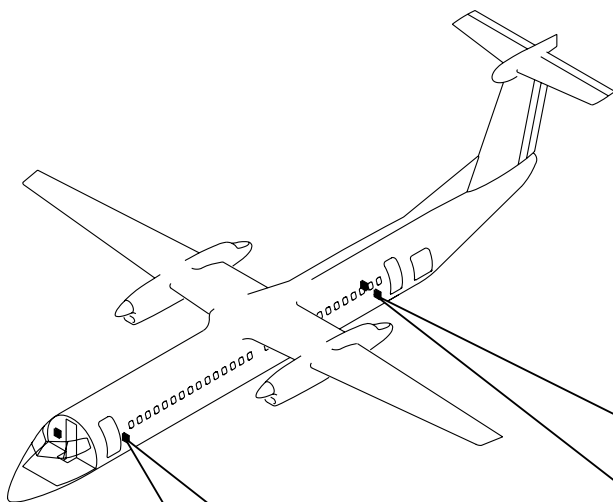
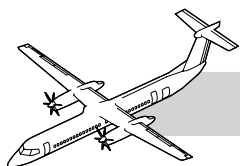
Refer to Figure 35-27. Passenger Emergency Oxygen System.

The passenger emergency (First Aid) oxygen system is intended for two persons use. It has the masks connected by the hoses to the outlets of an oxygen bottle.

The bottle components and operation is similar to Flight Attendant Portable Oxygen. There are four units installed in the cabin as shown in the adjacent graphic.

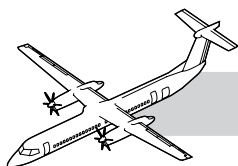
Servicing of the Crew, Passenger and First Aid Portable Oxygen Cylinders

All portable oxygen cylinders must be removed from the aircraft for charging.



PBE and Container

Figure 35-28. Portable Breathing Equipment Units (PBE)



Protective Breathing Equipment (PBE)

Refer to:

- Figure 35-28. Portable Breathing Equipment Units (PBE).
- Figure 35-29. Passenger Compartment Emergency Equipment.

The PBE is a self contained, portable, disposable breathing device. The system will supply oxygen for approximately 15 minutes. The PBE has a 10 year useful life.

The unit is kept in a vacuum sealed pouch for protection from moisture and contamination. The unit is ready for use as long as the pull strip on the sealed pouch is intact and there is no visual indication of loss of vacuum.

Each PBE unit has a hood which seals around the wearer's neck to keep out any harmful gases and fumes. The hood has a clear view front panel which allows for aural communication.

There are four PBE Units installed on the aircraft. One is located in the flight compartment in a storage container attached to the bulkhead behind the co-pilot seat and three are located in the cabin near the fire extinguishers.

SYSTEM OPERATION

Oxygen is activated by sharply snapping the two oxygen cylinders away from each other. After activating the cylinders, the hood must be donned as quickly as possible, taking no longer than 10 seconds.

One cylinder will release oxygen rapidly and an audible hissing sound indicates that the PBE is operating properly.

The second cylinder discharges oxygen more slowly. A flashing green in the service indicator shows the crew member that the low-flow cylinder is operating properly.

Once donned, the neck seal acts as a relief valve to prevent excessive pressure from building inside the hood.

NOTE

Do not use the PBE if the tamper-evident seals are broken, torn or missing.

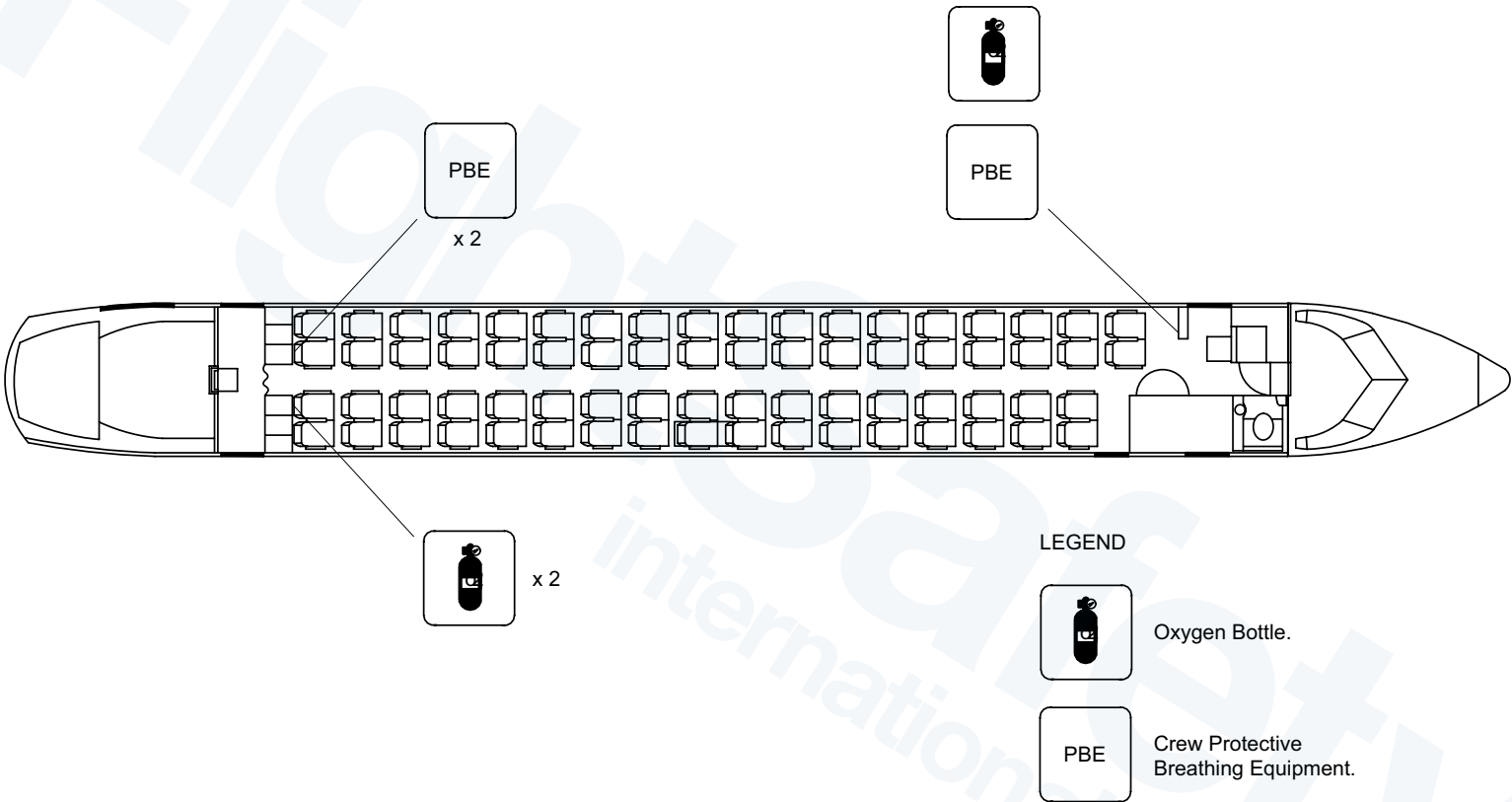
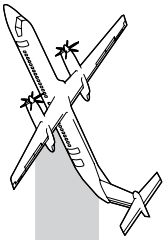
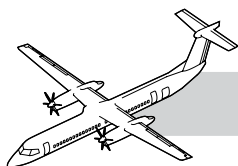


Figure 35-29. Passenger Compartment Emergency Equipment



35-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB3500001 Connector - Oxygen Outlet (Purging Tool)
- GSB5180001 Tester - Electrical Bonding (Portable DLRO)
- None specified Nitrogen Test Rig
- GSB1216012 Nitrogen rig, low pressure
- GSB2525002 Tool - PSU Mounting Wheel
- CDSP4644-501 Manual deployment oxygen tool

35-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 35-10-00-617-801: Flush the Crew Oxygen System.
- AMM 12-10-35-610-801: Servicing of the Crew Oxygen System - 39.8 cu. ft. Cylinder.
- AMM 12-10-35-610-802: Servicing of the Crew Oxygen System - 50 cu. ft. Cylinder.
- AMM 35-10-00-710-801: Operational Test of the Crew Oxygen System.
- AMM 35-20-00-710-801: Operational Test of the Passenger Oxygen System.
- AMM 35-20-00-710-802: Operational Test of the Passenger Oxygen System Manual-Deployment Function.
- AMM 35-20-00-710-803: Operational Test of the Passenger Oxygen System Automatic-Deployment Function (Flight Mode).
- AMM 35-26-00-720-801: Functional Check of the APPOS Barometric Switch (MRB#354000-205).
- AMM 35-36-01-000-801: Removal of the Protective Breathing Equipment.
- AMM 35-36-01-400-801: Installation of the Protective Breathing Equipment.